

Bitvector Rank and Select in succinct space

Jacobson's rank

(Note: the following assumes we are doing $B \cdot \text{rank}_1$ queries, but $B \cdot \text{rank}_0$ queries are also doable with same methods.)

Basic ideas:

When a bitvector is sparse enough, we can simply store answers for all 1-bits

When a bitvector is short enough, we can store all answers for all possible vectors and queries

Jacobson's rank



Jacobson's rank



Split the string into chunks of length $\log_2^2 n$

Jacobson's rank



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Jacobson's rank



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Jacobson's rank



Split the string into chunks of length $\log_2^2 n$



$$\log_2^2 n \equiv (\log_2 n)^2$$

Jacobson's rank

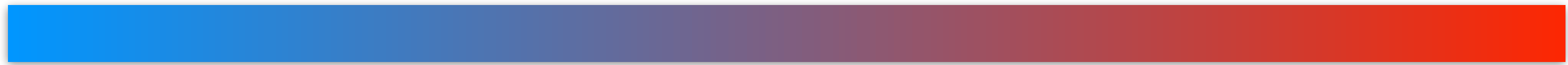


Split the string into chunks of length $\log_2^2 n$



$\log_2^2 n \equiv (\log_2 n)^2$ I'll omit base-2 from logs from now on

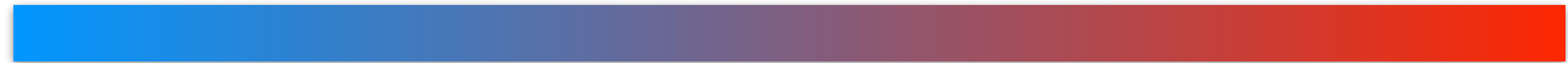
Jacobson's rank



$\lceil \log^2 n \rceil$

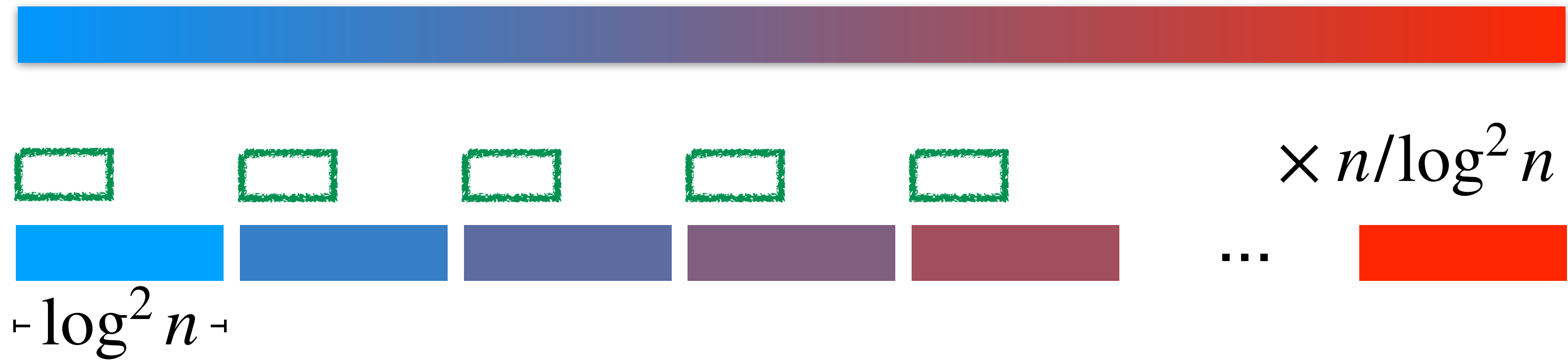
$\times n/\log^2 n$
...

Jacobson's rank



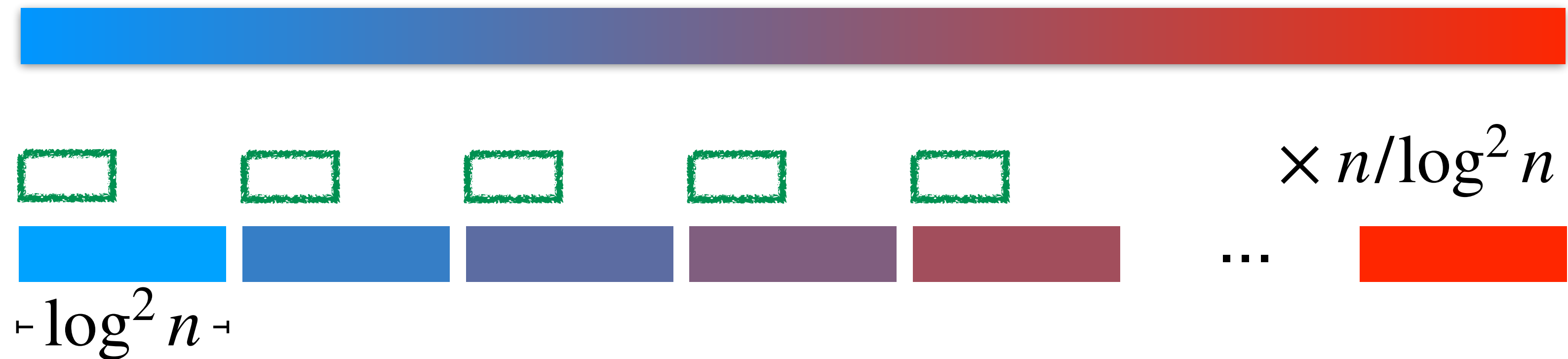
Store pre-calculated **cumulative rank** up to each chunk

Jacobson's rank



Store pre-calculated **cumulative rank** up to each chunk

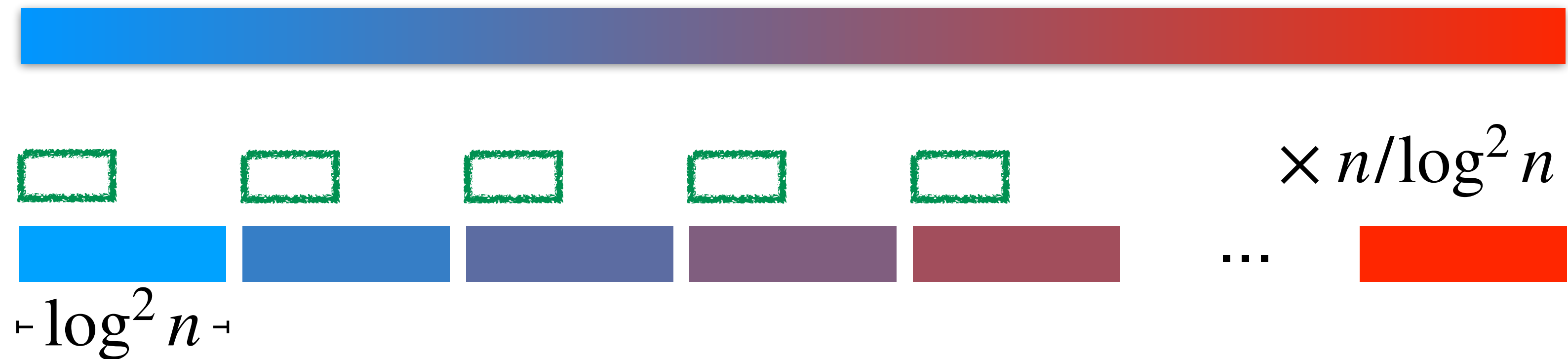
Jacobson's rank



Store pre-calculated **cumulative rank** up to each chunk

$$O(\log n \cdot n/\log^2 n) = O(n/\log n) = \check{O}(n)$$

Jacobson's rank

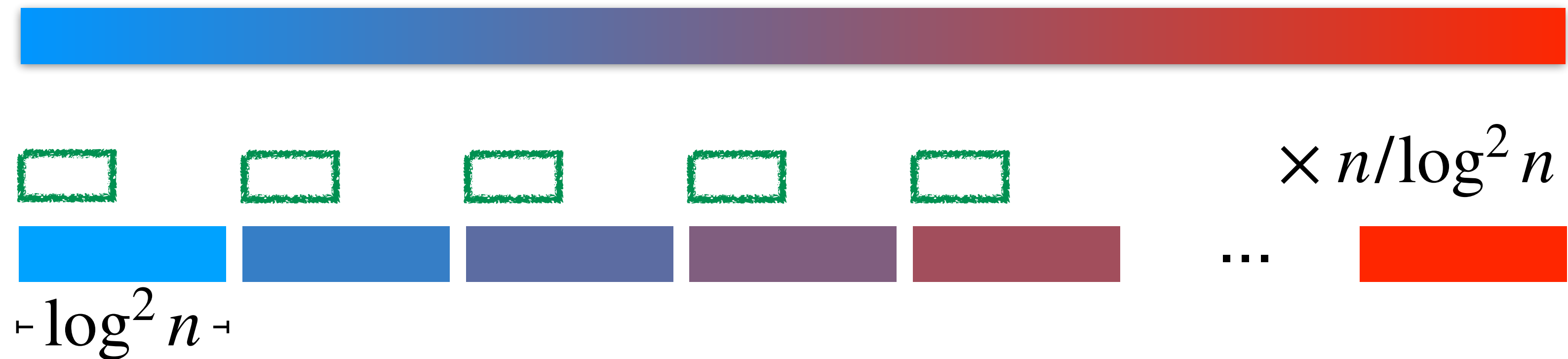


Store pre-calculated **cumulative rank** up to each chunk

$$O\left(\underbrace{\log n}_{\text{bits to store cum. rank}} \cdot n/\log^2 n\right) = O\left(n/\log n\right) = \check{O}(n)$$

bits to store
cum. rank

Jacobson's rank



Store pre-calculated **cumulative rank** up to each chunk

$$O \left(\underbrace{\log n}_{\text{bits to store cum. rank}} \cdot \underbrace{n/\log^2 n}_{\# \text{ chunks}} \right) = O \left(n/\log n \right) = \tilde{o}(n)$$

Jacobson's rank



So far, extra space is
 $\tilde{O}(n)$



Jacobson's rank



So far, extra space is
 $\tilde{O}(n)$

Finding a rank can be decomposed:



Jacobson's rank



So far, extra space is
 $\tilde{O}(n)$

Finding a rank can be decomposed:

(a) find what chunk it's in (division)



Jacobson's rank



So far, extra space is $\tilde{O}(n)$

Finding a rank can be decomposed:

(a) find what chunk it's in (division)

(b) look up **cumulative rank**



Jacobson's rank



So far, extra space is $\tilde{O}(n)$

Finding a rank can be decomposed:

(a) find what chunk it's in (division)

(b) look up **cumulative rank**

(c) find (relative) rank *within* chunk



Jacobson's rank



So far, extra space is $\tilde{O}(n)$

Finding a rank can be decomposed:

(a) find what chunk it's in (division)

(b) look up **cumulative rank**

(c) find (relative) rank *within* chunk

(d) add (b) + (c)



Jacobson's rank



So far, extra space is $\tilde{O}(n)$

Finding a rank can be decomposed:

(a) find what chunk it's in (division)

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TODO



Jacobson's rank

chunk



$$-\log^2 n -$$

Jacobson's rank

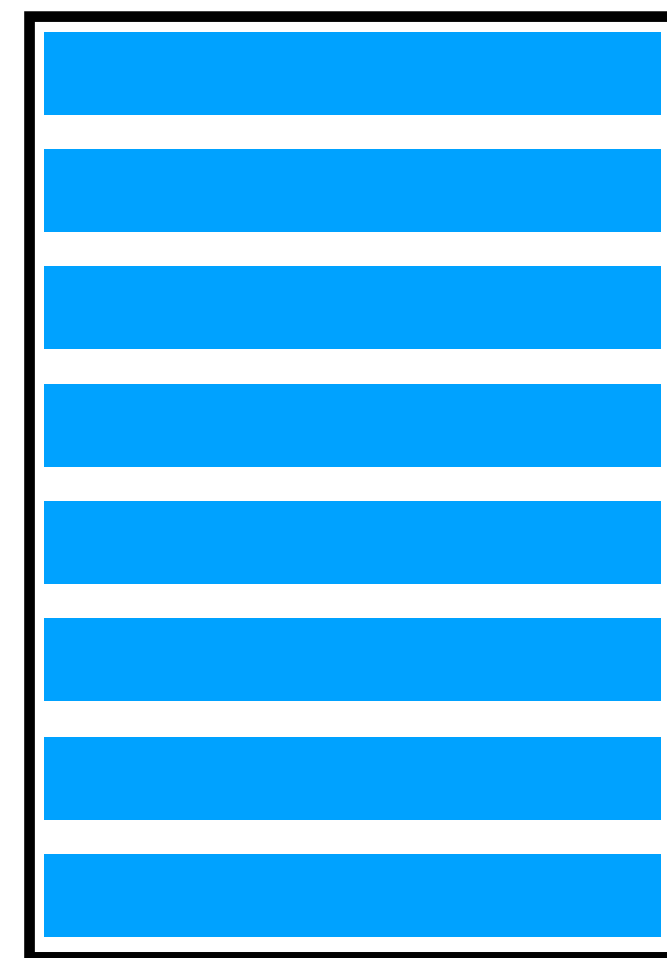
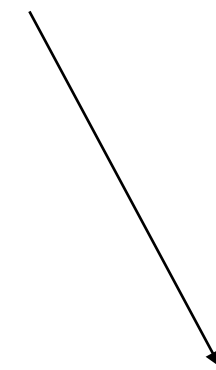


Say a chunk consists of $2 \log n$
sub-chunks, each of $1/2 \log n$
bits

Jacobson's rank



$\pm \log^2 n$



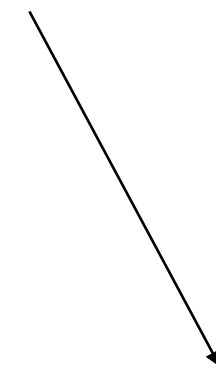
chunk

Say a chunk consists of $2 \log n$ sub-chunks, each of $1/2 \log n$ bits

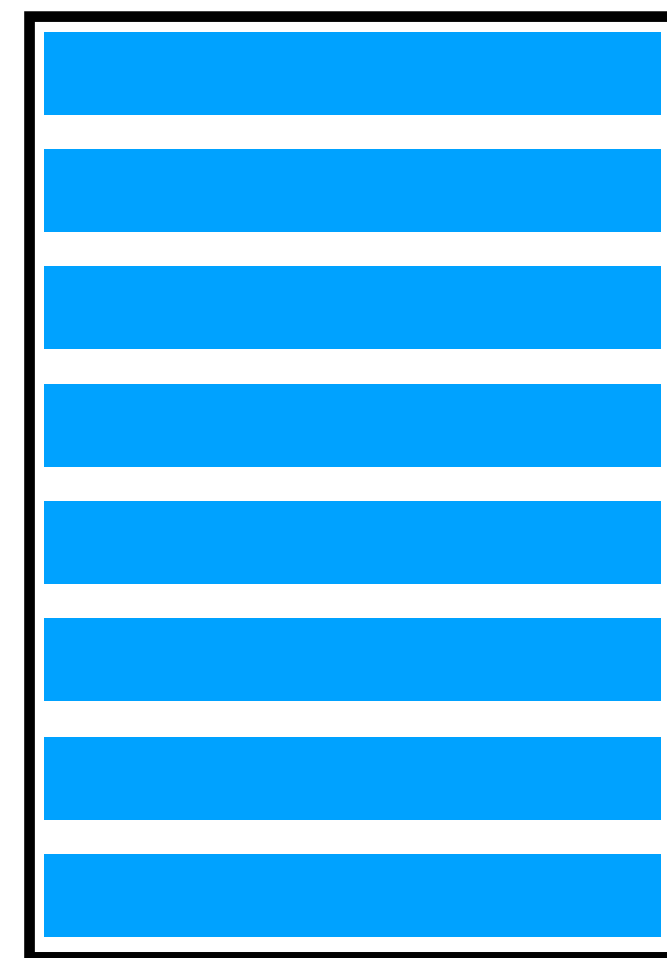
Jacobson's rank



$\pm \log^2 n$

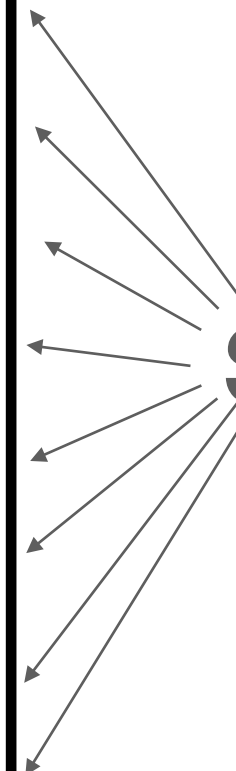


Say a chunk consists of $2 \log n$ sub-chunks, each of $1/2 \log n$ bits

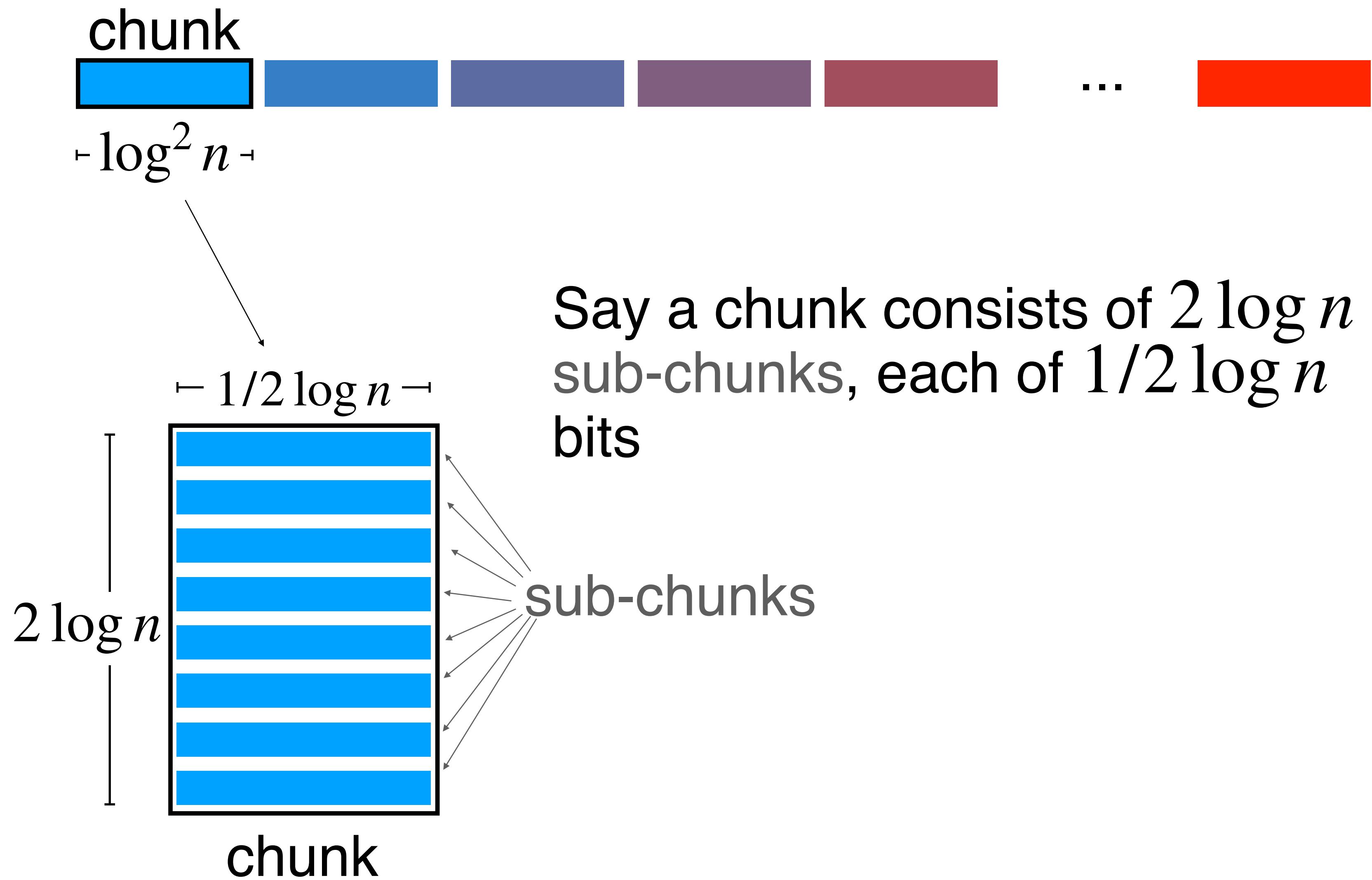


chunk

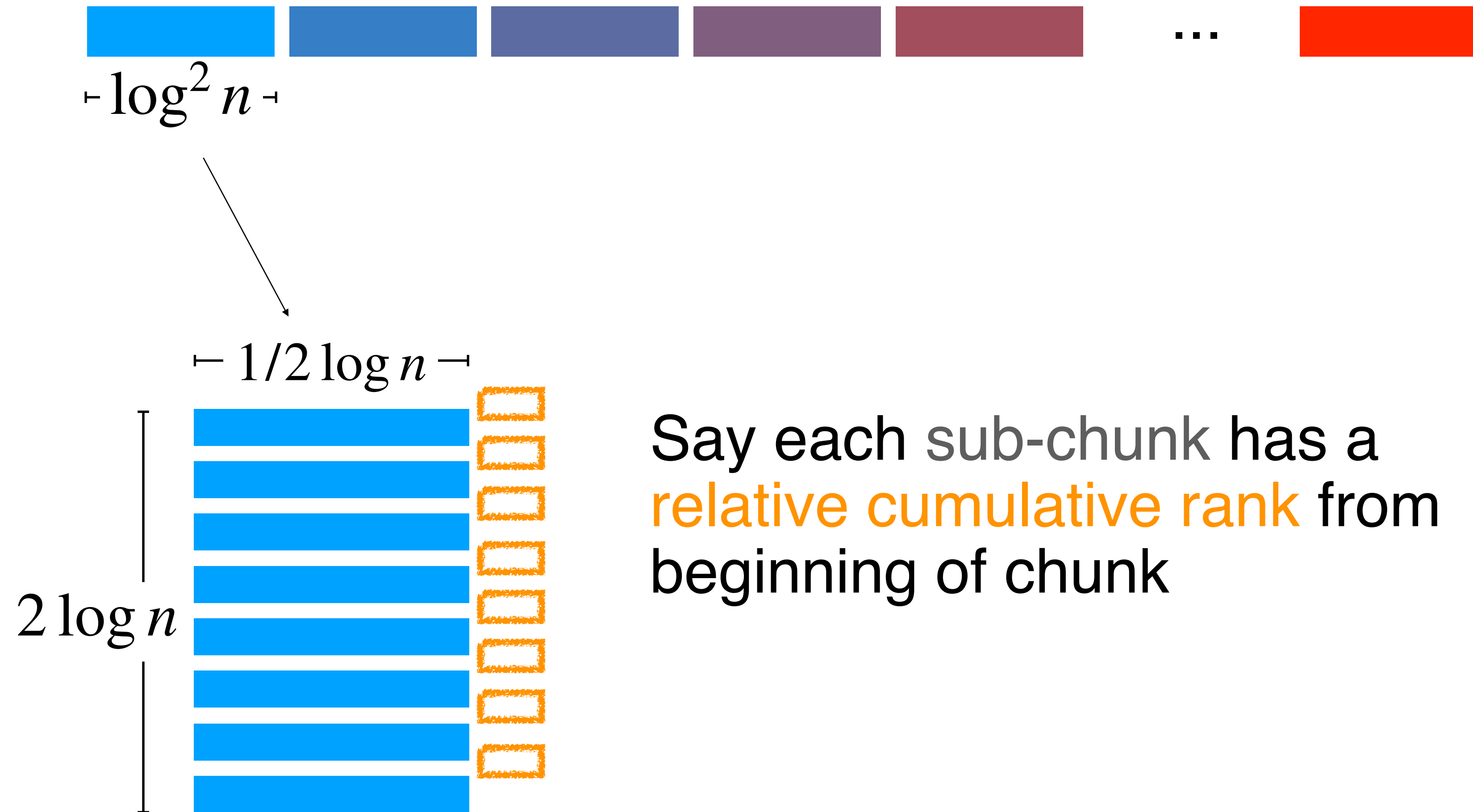
sub-chunks



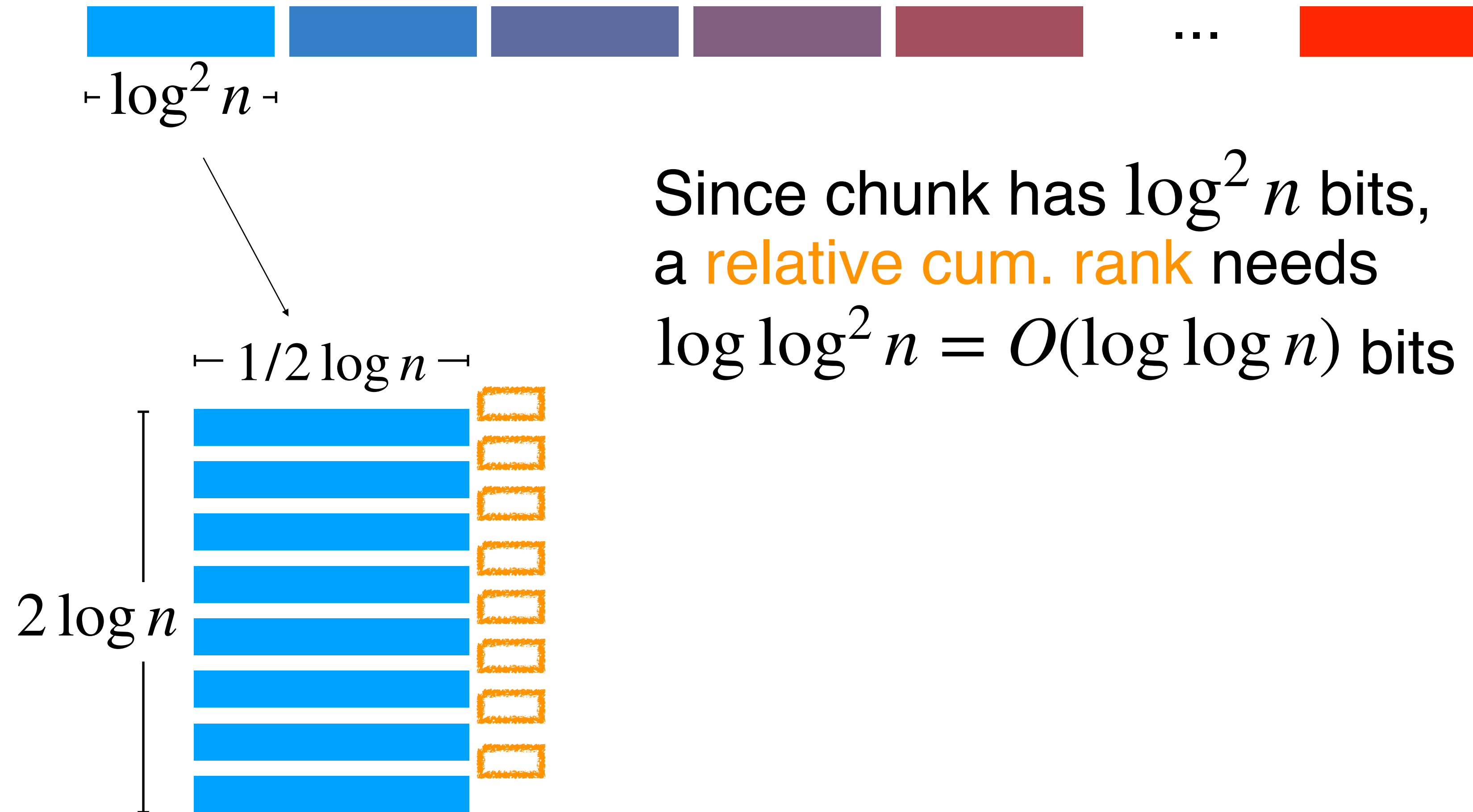
Jacobson's rank



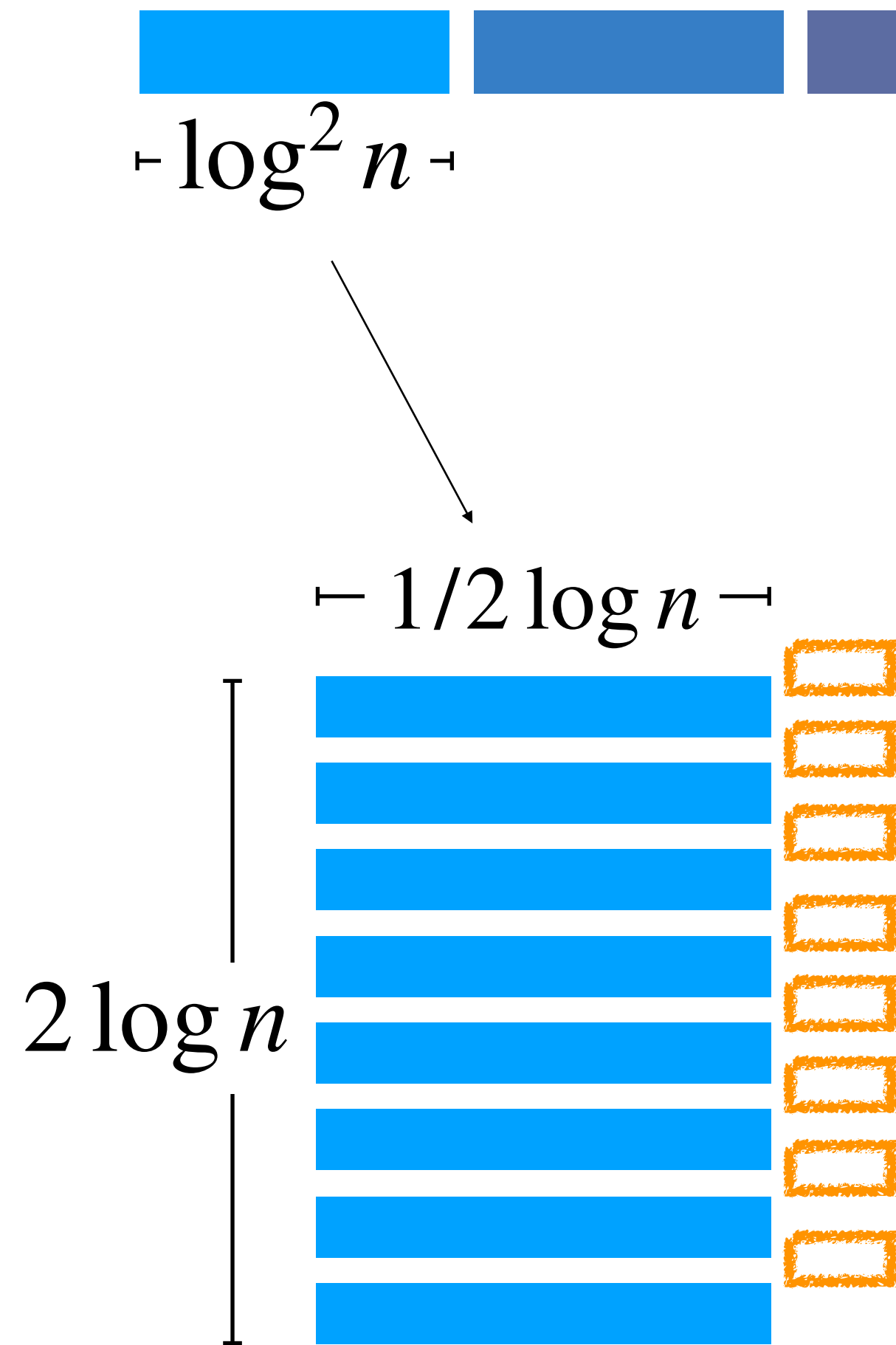
Jacobson's rank



Jacobson's rank



Jacobson's rank

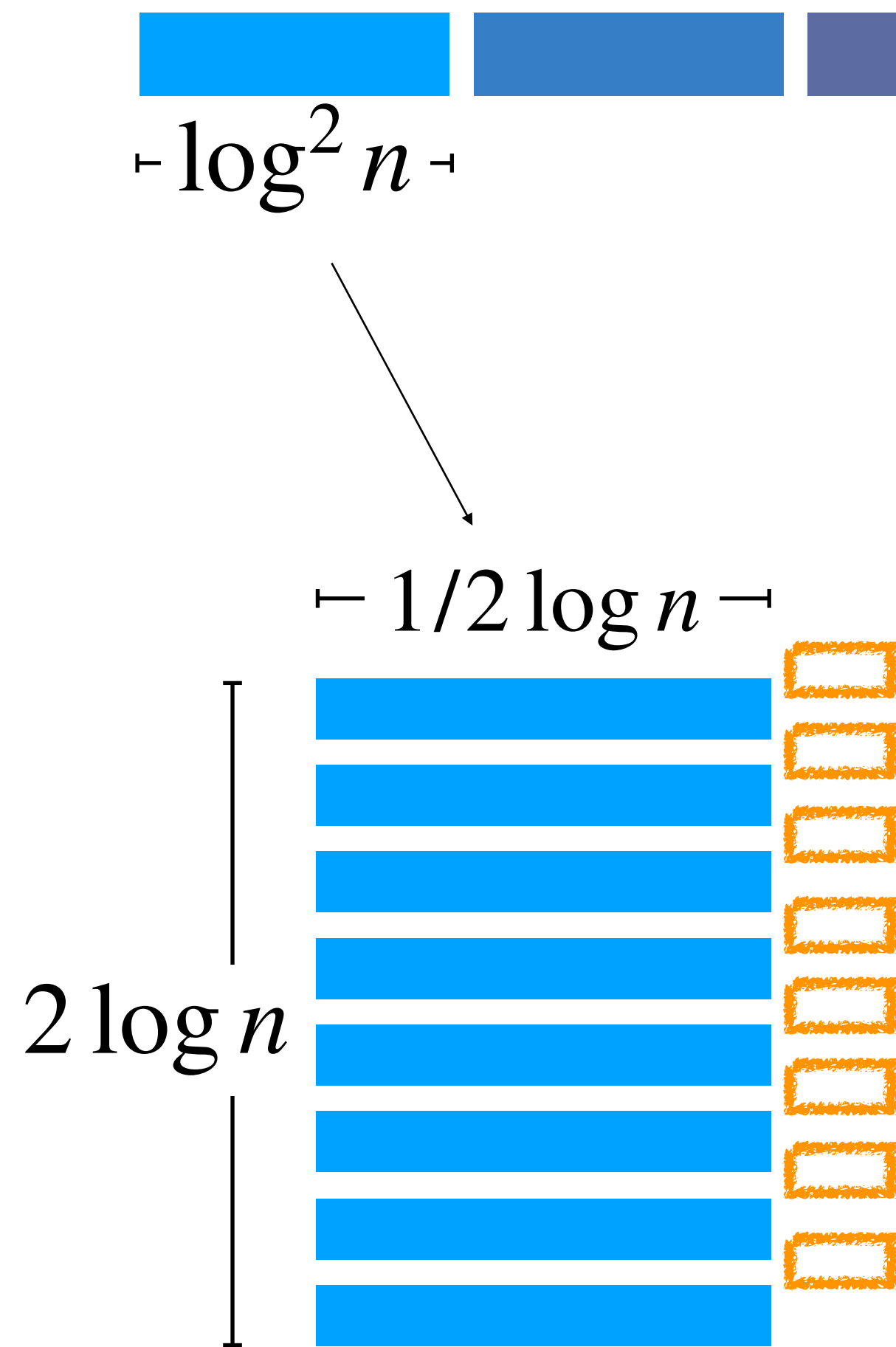


Since chunk has $\log^2 n$ bits,
a **relative cum. rank** needs
 $\log \log^2 n = O(\log \log n)$ bits

$O(n/\log n)$ sub-chunks overall
(across all chunks),

for total of
 $O(n \cdot \log \log n / \log n)$ bits for
relative cum. ranks

Jacobson's rank



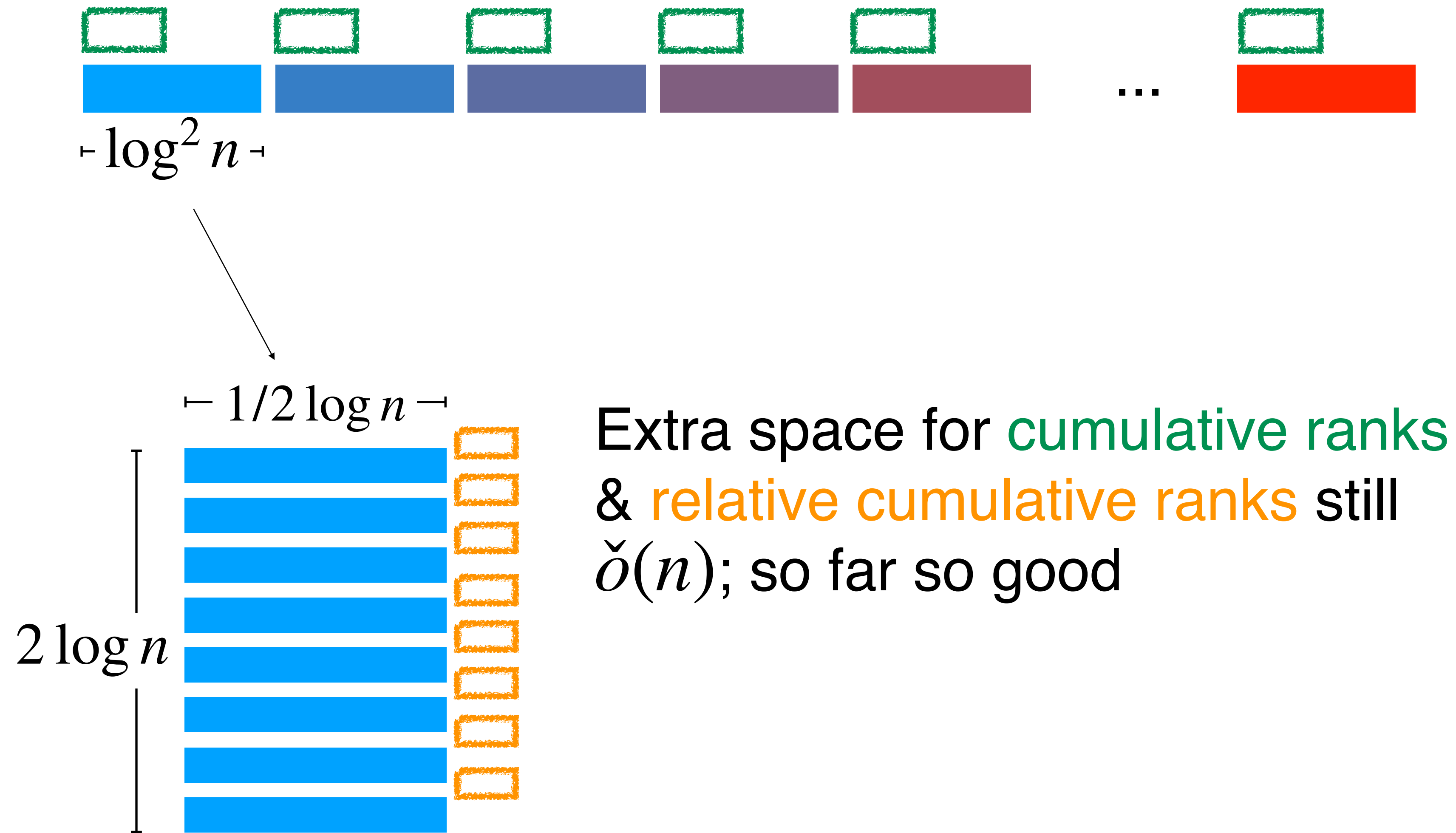
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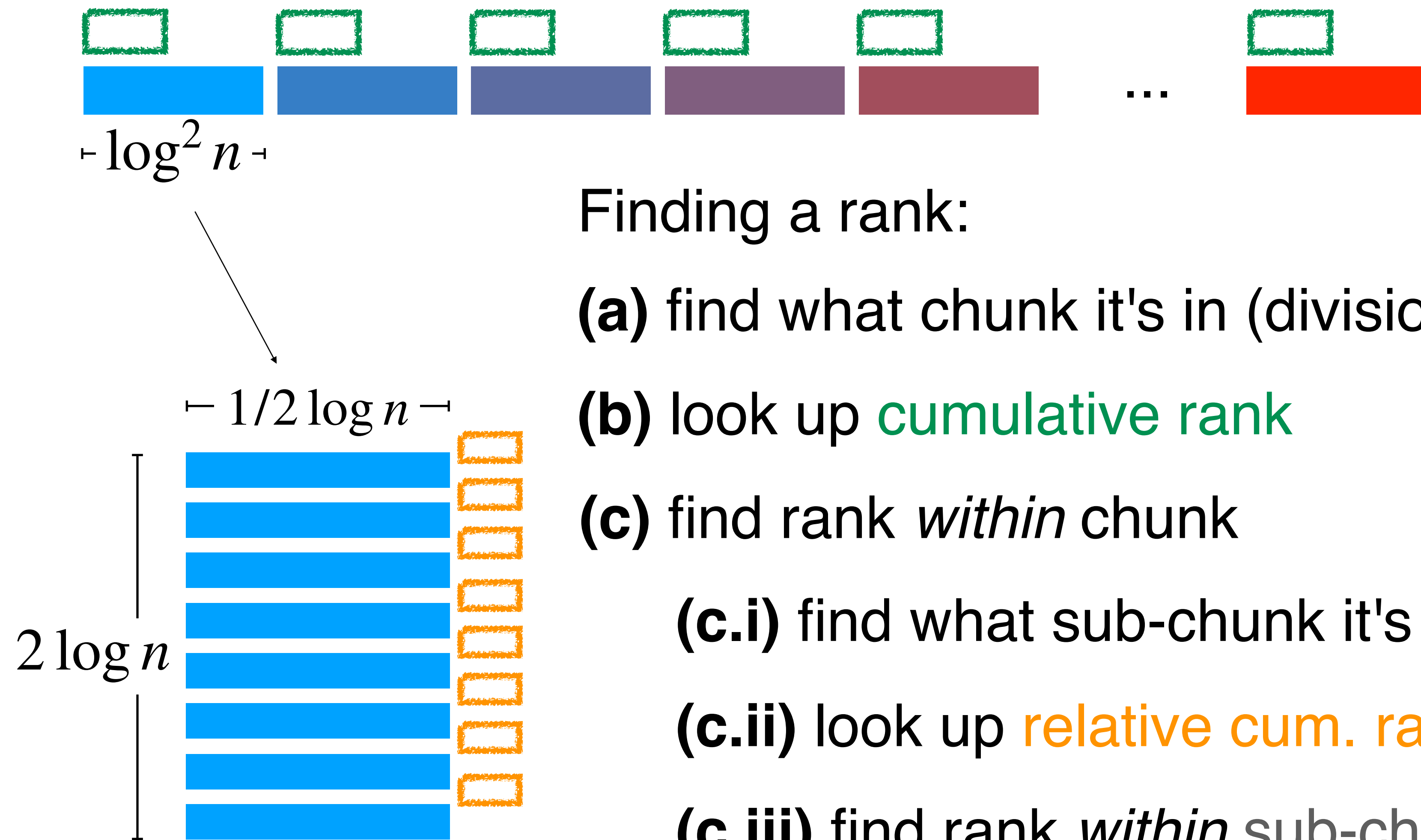
$$O(n \cdot \log \log n / \log n) = \tilde{o}(n)$$

Jacobson's rank



Extra space for **cumulative ranks**
& **relative cumulative ranks** still
 $\tilde{O}(n)$; so far so good

Jacobson's rank



Finding a rank:

(a) find what chunk it's in (division)

(b) look up **cumulative rank**

(c) find rank *within* chunk

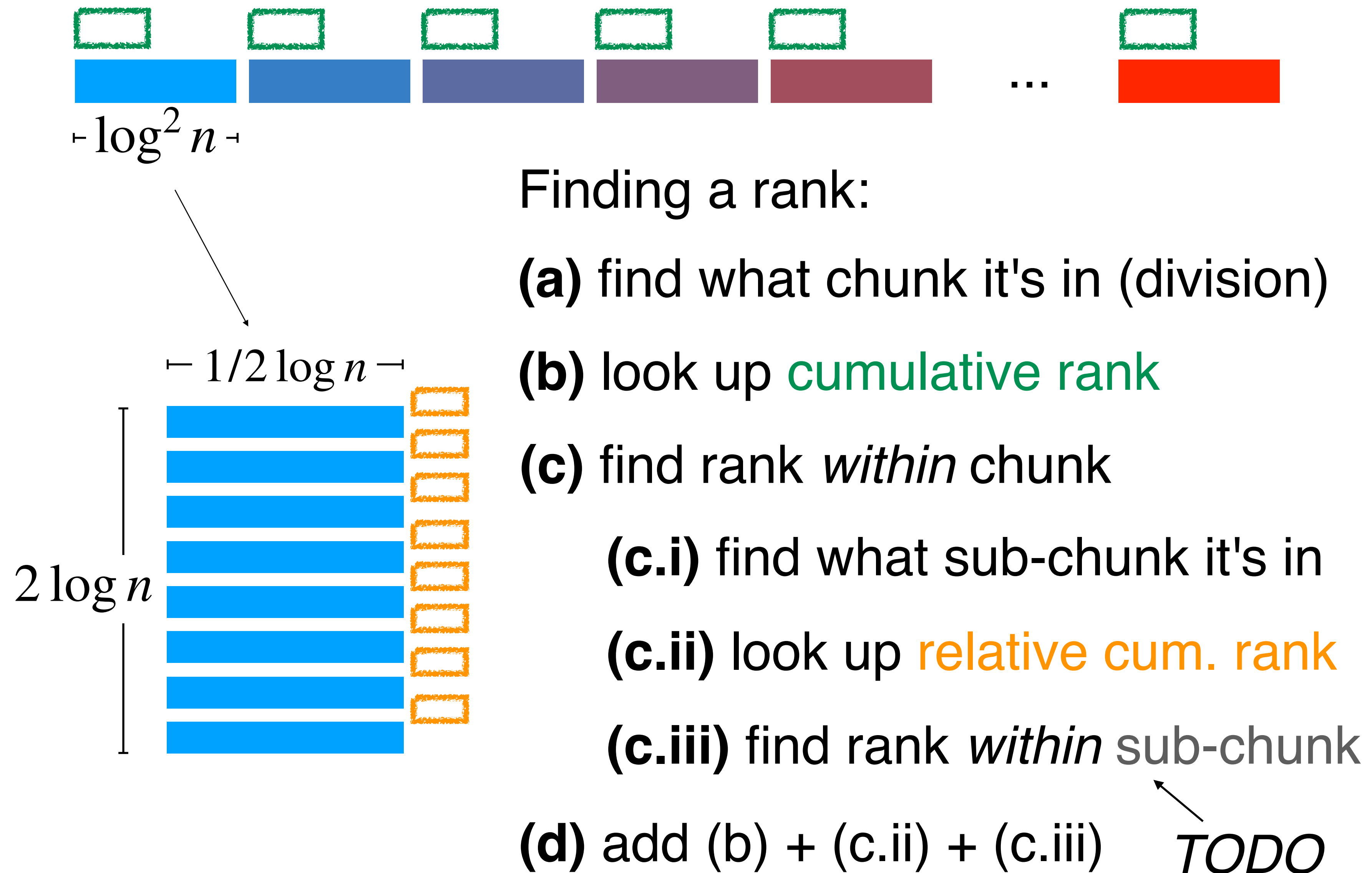
(c.i) find what sub-chunk it's in

(c.ii) look up **relative cum. rank**

(c.iii) find rank *within* sub-chunk

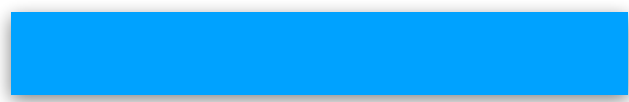
(d) add (b) + (c.ii) + (c.iii)

Jacobson's rank



Jacobson's rank

$\lceil 1/2 \log n \rceil$



Finding rank *within a sub-chunk*:
two ways of thinking

Jacobson's rank

⌊ $1/2 \log n$ ⌋

Finding rank *within a sub-chunk*:
two ways of thinking

Way 1: $1/2 \log n$ is $\sim$ a machine word; use instructions like "population count" to find rank in $O(1)$ time

Jacobson's rank

⌊ $1/2 \log n$ ⌋

Finding rank *within a sub-chunk*:
two ways of thinking

Way 1: $1/2 \log n$ is $\sim$ a machine word; use instructions like "population count" to find rank in $O(1)$ time

Way 2: Lookup table

(Next slide)

Jacobson's rank

Say we naively store answers to all rank queries for all length- x bitvectors. How many bits required?

$$2^x$$

possible
bitvectors

Jacobson's rank

Say we naively store answers to all rank queries for all length- x bitvectors. How many bits required?

$$2^x \cdot x$$

possible possible
bitvectors offsets

Jacobson's rank

Say we naively store answers to all rank queries for all length- x bitvectors. How many bits required?

$$2^x \cdot x \cdot \log x$$

possible possible answer
bitvectors offsets

Jacobson's rank

Say we naively store answers to all rank queries for all length- x bitvectors. How many bits required?

$$2^x \cdot x \cdot \log x$$

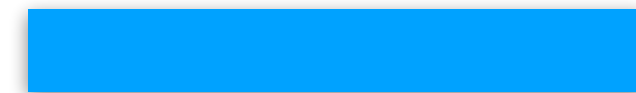
possible
bitvectors

possible
offsets

answer

sub-chunk

$\lceil 1/2 \log n \rceil$



Let $x = 1/2 \log n$

Jacobson's rank

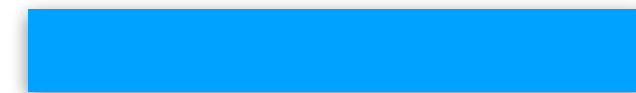
Say we naively store answers to all rank queries for all length- x bitvectors. How many bits required?

$$2^x \cdot x \cdot \log x$$

possible possible answer
bitvectors offsets

sub-chunk

$$\lceil 1/2 \log n \rceil$$



$$\text{Let } x = 1/2 \log n$$

$$\begin{aligned} 2^{1/2 \log n} \cdot 1/2 \log n \cdot \log 1/2 \log n &= O\left(\sqrt{n} \log n \log \log n\right) \\ &= \tilde{o}(n) \end{aligned}$$

Storing an “all query” lookup table

- Blocks have size $b = \log_2 n / 2$
 - There are 2^b such blocks possible
 - In each block there are b possible rank queries
 - Each answer (relative to the block) is in the range $1..b$

R_p	Type	0	1	rank(0)	rank(1)
	0	0	0	0	0
	1	0	1	0	1
	2	1	0	1	1
	3	1	1	1	2

- Therefore size of R_p , the in-block data structure is
 - $2^b * b * \log b = n^{1/2} * \log n * \log \log n / 2$ bits = $o(n)$ bits

Jacobson's rank

Finding a rank:

- (a)** find what chunk it's in (division)
- (b)** look up **checkpoint**
- (c)** find rank *within* chunk
 - (c.i)** find what sub-chunk it's in
 - (c.ii)** look up **relative checkpoint**
 - (c.iii)** find rank *within* sub-chunk
- (d)** add (b) + (c.ii) + (c.iii)

Jacobson's rank

Finding a rank:

(a) find what chunk it's in (division)

(b) look up **checkpoint**

(c) find rank *within* chunk

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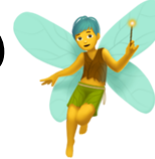
(d) add (b) + (c.ii) + (c.iii)

} $O(1)$

Bitvectors

	Time	Space (bits)	Note
$B . \text{access}$	$O(1)$	n	Lookup
$B . \text{select}_1$	$O(1)$	$\tilde{O}(n)$	
$B . \text{rank}_1$	$O(1)$	$\tilde{O}(n)$	 Jacobson

Bitvectors

	Time	Space (bits)	Note
$B . \text{access}$	$O(1)$	n	Lookup
$B . \text{select}_1$	$O(1)$	$\tilde{O}(n)$	?    ?
$B . \text{rank}_1$	$O(1)$	$\tilde{O}(n)$	✓ Jacobson

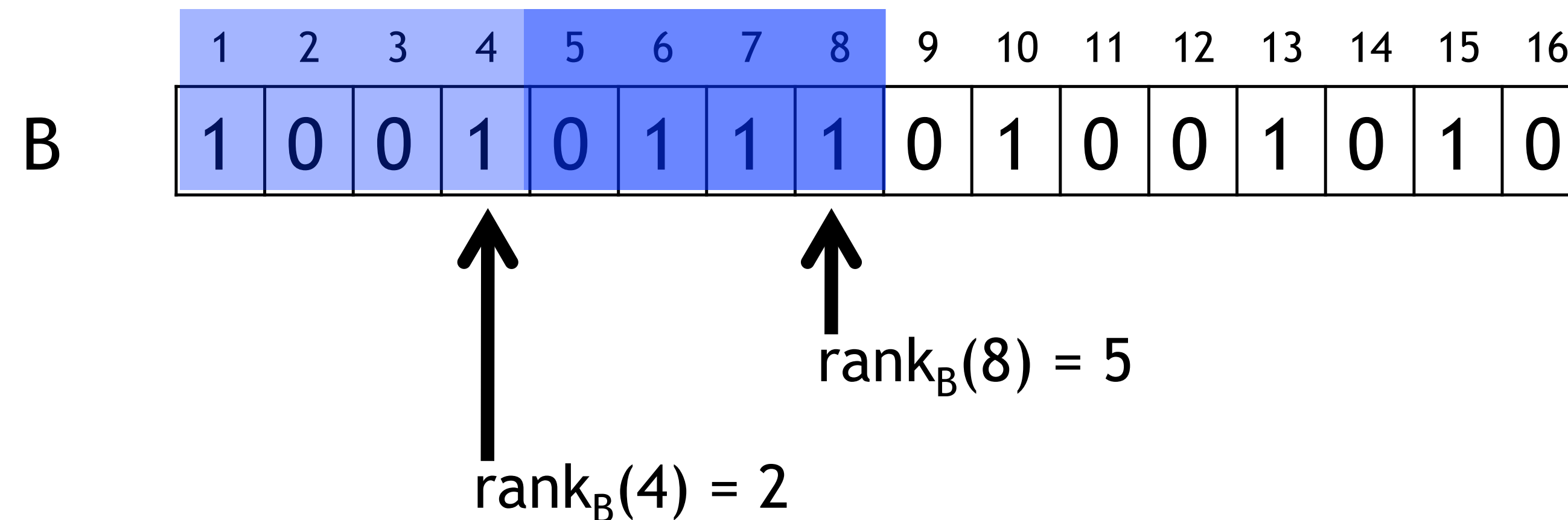
More details on rank

- Rank index takes $O(n \log \log n / \log n) = o(n)$ bits so we use $n + o(n)$ overall and can answer queries in $O(1)$ time
- While it is sublinear, we'd still like the $o(n)$ term to be small
 - Best is by Patrascu: $O(n / \log^k n)$ bits, $O(k)$ time queries
- Dynamic solutions exist
 - Queries no longer constant: $O(\log n / \log \log n)$ time (Raman et al.)

Select in $O(\lg n)$ via rank

- We can use our solution to rank to get a (fairly) efficient solution to $\text{select}(i)$, with this observation:
- If $\text{rank}(n/2) > i$, then the i^{th} 1-bit is in $B[1..n/2]$
 - Otherwise it is in $B[n/2+1..n]$

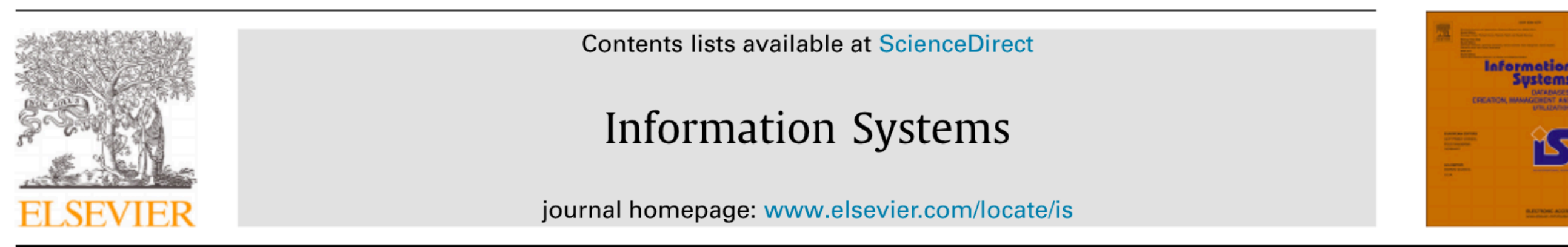
$\text{select}_B(3)$



Other solutions for rank & select

- Applying this idea recursively to arrive at select(i)
 - $O(\log_2 n)$ time, $o(n)$ space
- $O(1)$ time, $o(n)$ space solutions for select also exist
 - Slightly more complicated than $O(1)$ rank
 - (Munro and Clark)
- Similar variations as we discussed with rank (trading space for query time) are also possible

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Rank and select: Another lesson learned

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Clark's select

Clark's select

Unlike rank:

Clark's select

Unlike rank:

Chunks are defined by # 1s, not # bits

Clark's select

Unlike rank:

Chunks are defined by # 1s, not # bits

Two layers of “sparsity”

Clark's select

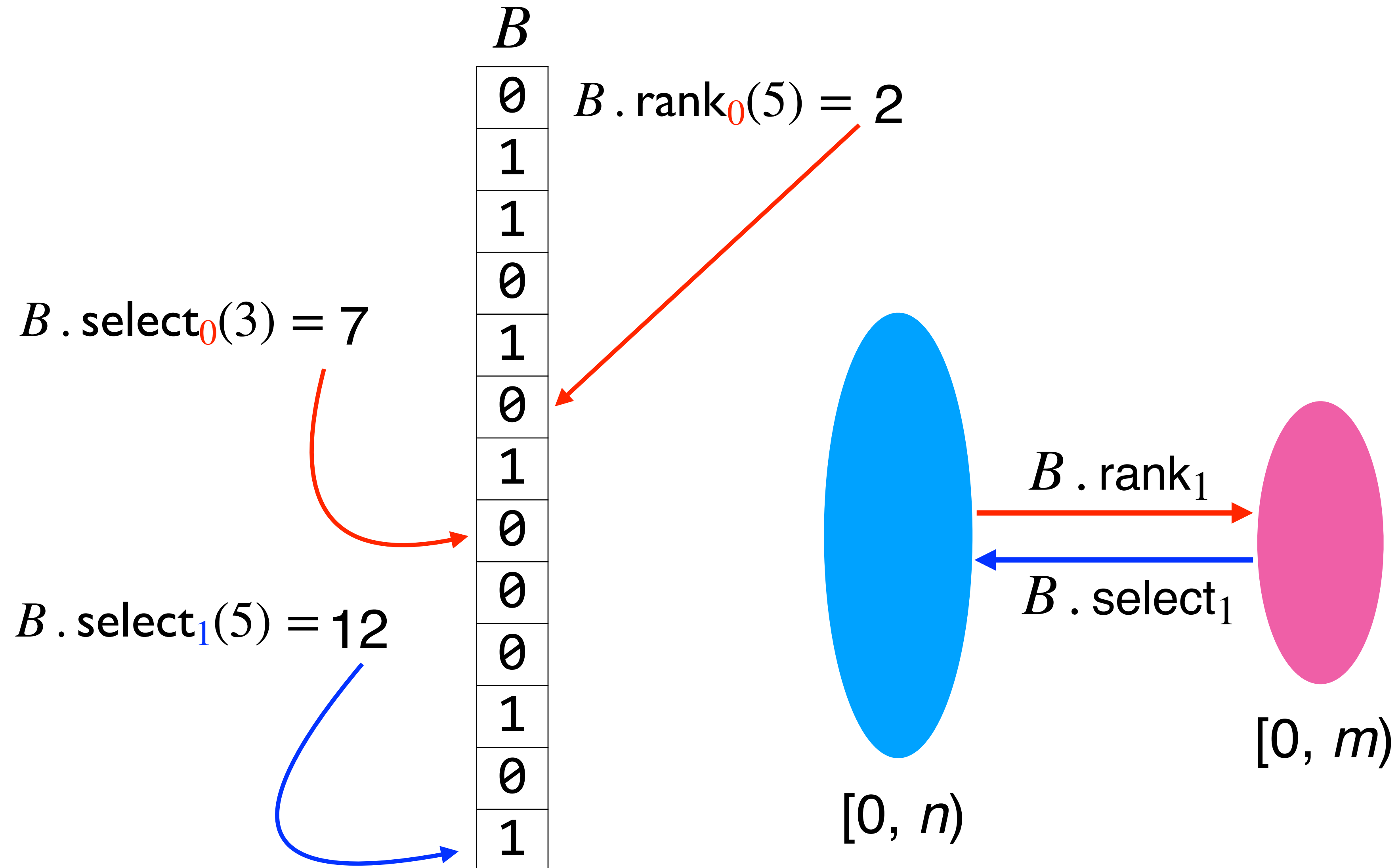
Unlike rank:

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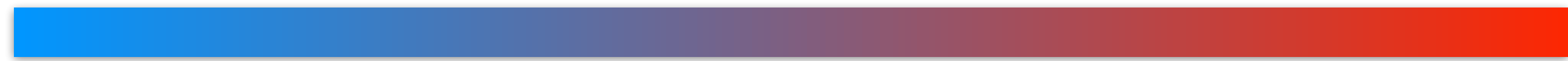
Two layers of “sparsity”

Answer is an offset into bitvector

Bitvectors



Clark's select



Split into $\log^2 n$ -weight chunks

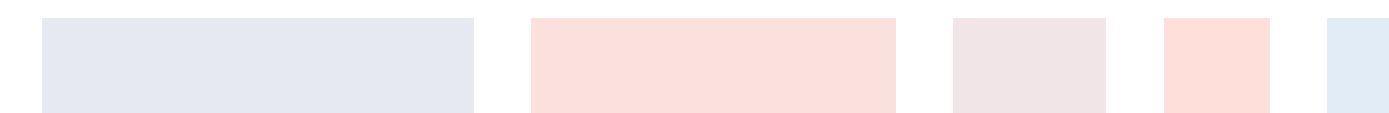


Sparse ($\geq \log^4 n$ -length)

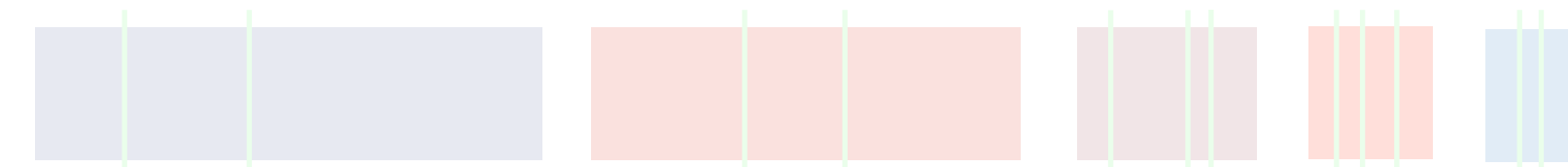


Lookup table for
each 1-bit ✓

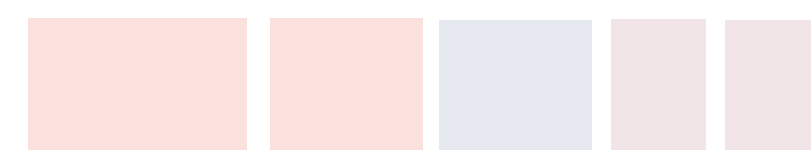
Dense ($< \log^4 n$ -length)



Split into $\sqrt{\log n}$ -weight sub-chunks

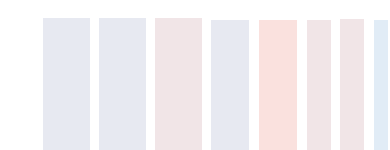


Sparse ($\geq 1/2 \log n$ -length)



Lookup table for
each 1-bit ✓

Dense ($< 1/2 \log n$ -length)



Lookup table for all
possible sub-chunks ✓

Clark's select



Split the string into chunks each containing
 $\log^2 n$ 1-bits

Clark's select



Split the string into chunks each containing
 $\log^2 n$ 1-bits



Clark's select



Split the string into chunks each containing
 $\log^2 n$ 1-bits



Larger chunks are sparse; 1's spread out

Clark's select



Split the string into chunks each containing
 $\log^2 n$ 1-bits



Larger chunks are sparse; 1's spread out

Shorter chunks are dense; 1's packed together

Clark's select

Each chunk contains $\log^2 n$ 1-bits



Worst case:
every bit set

Clark's select

Each chunk contains $\log^2 n$ 1-bits



We store **offset** of each chunk start



Worst case:
every bit set

Clark's select

Each chunk contains $\log^2 n$ 1-bits



We store **offset** of each chunk start



Worst case:
every bit set

Clark's select

Each chunk contains $\log^2 n$ 1-bits



We store **offset** of each chunk start

This takes:



Worst case:
every bit set

Clark's select

Each chunk contains $\log^2 n$ 1-bits



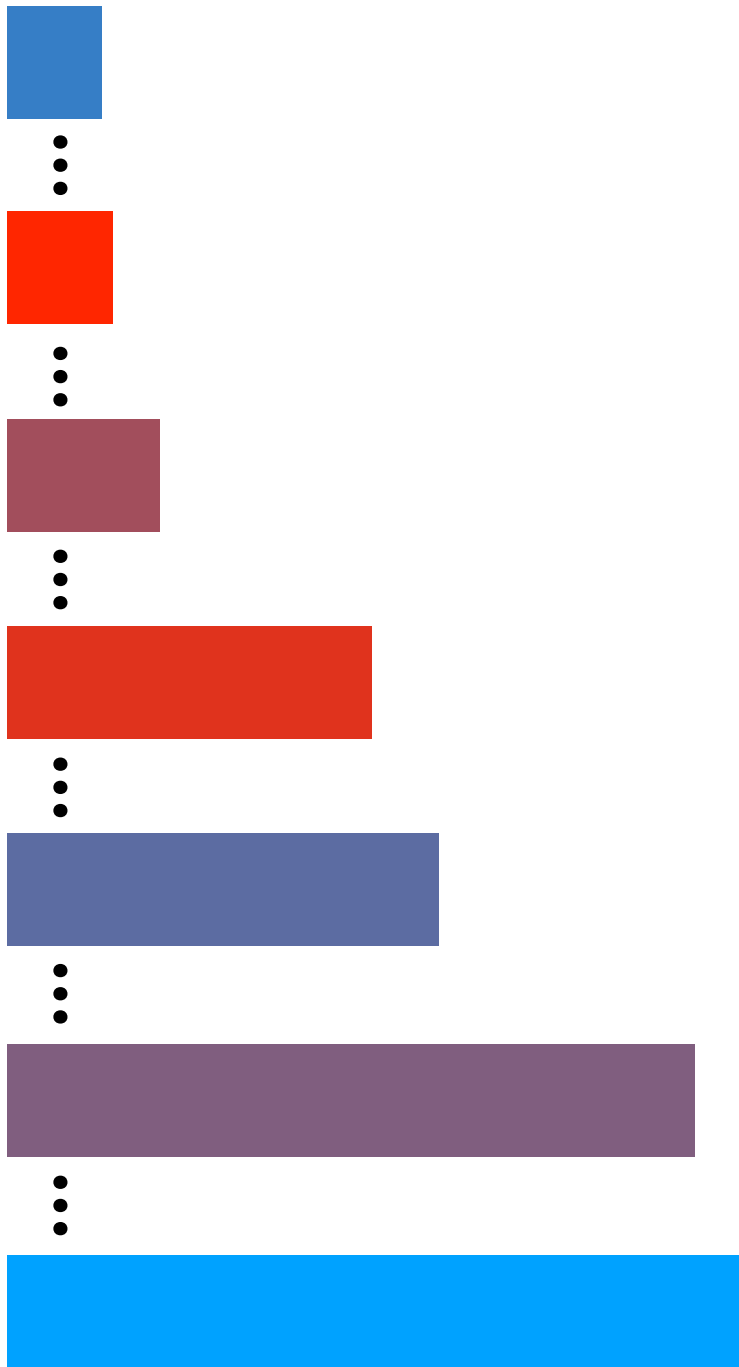
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This takes:

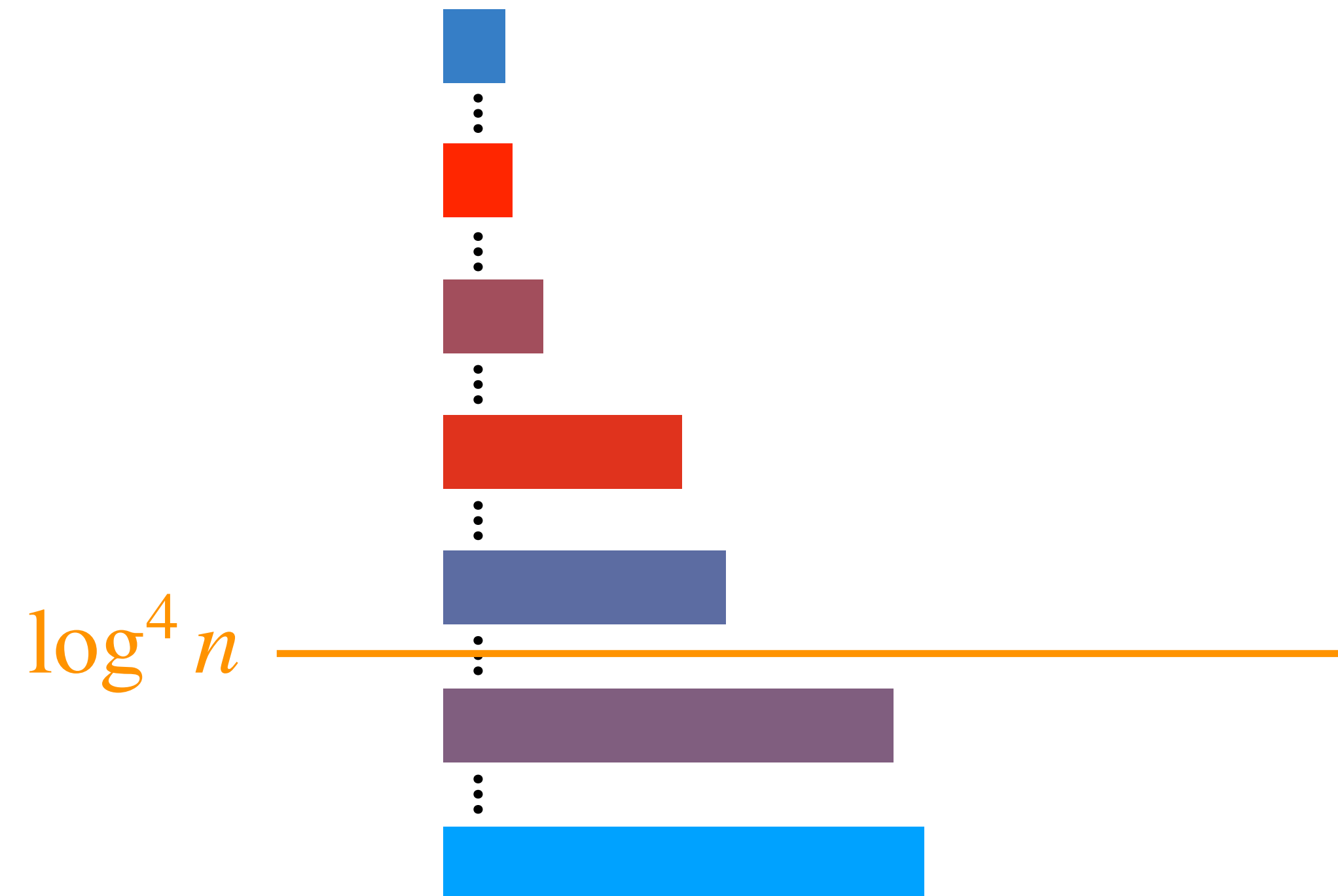
$$O\left(\underbrace{\frac{n}{\log^2 n} \log n}_{\text{Worst case: every bit set}}\right) = O\left(\frac{n}{\log n}\right) = \check{O}(n) \text{ bits}$$

Worst case:
every bit set

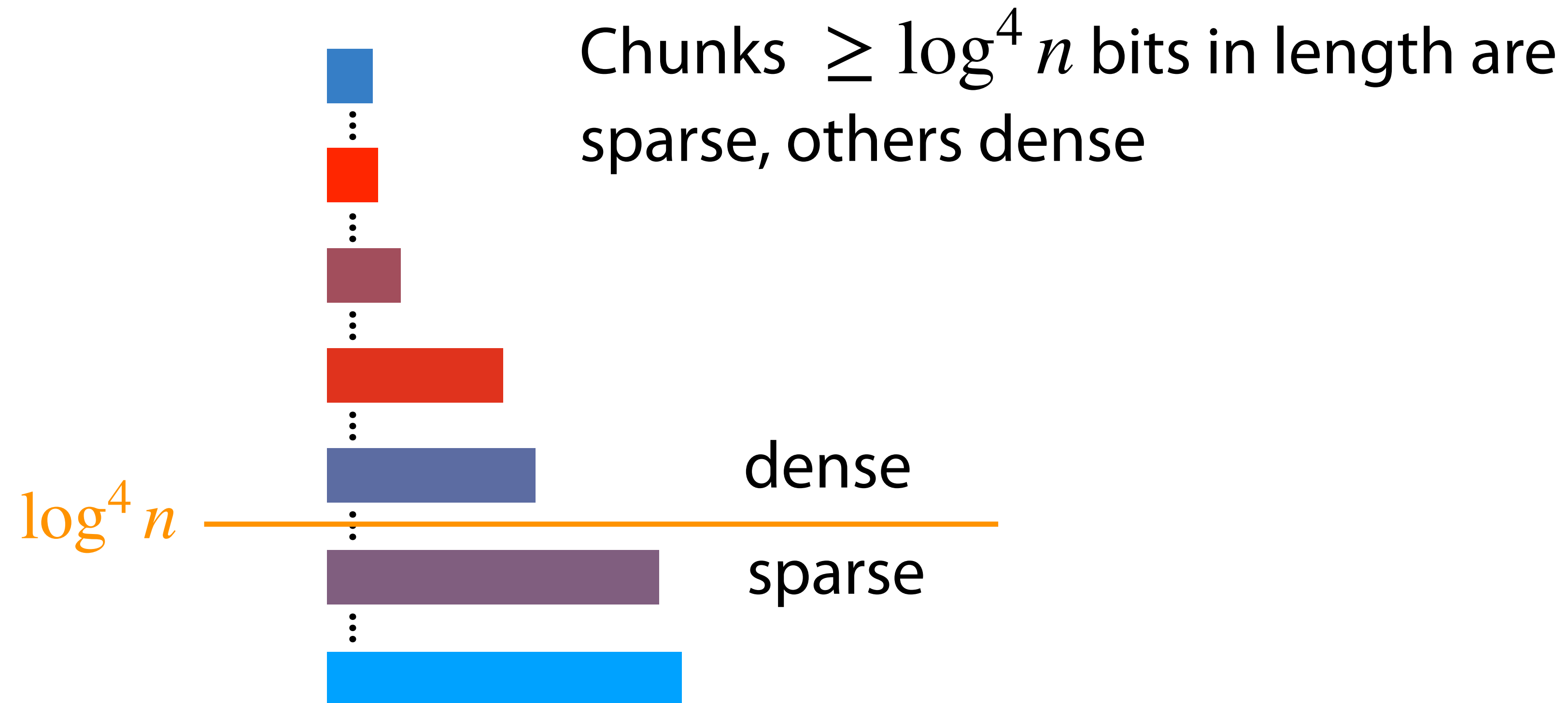
Clark's select



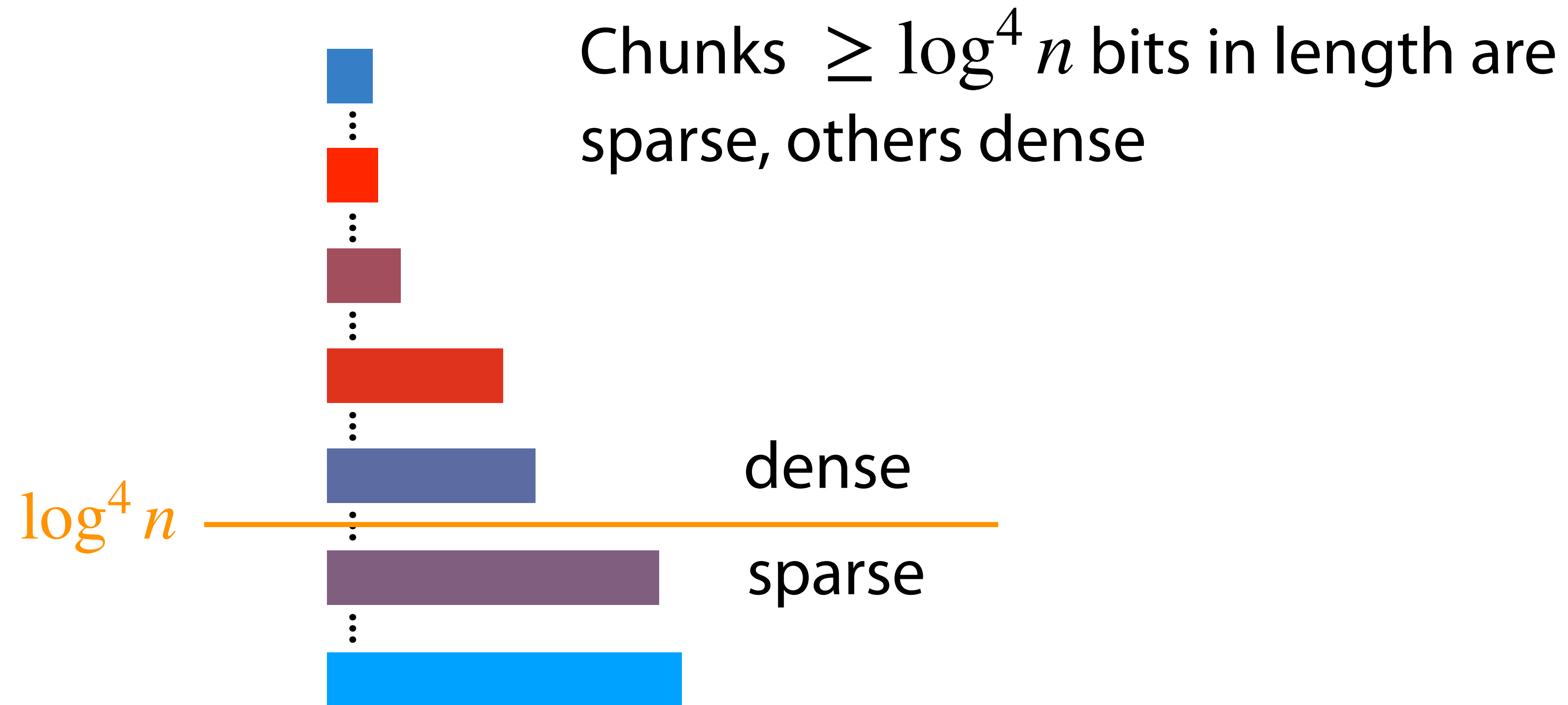
Clark's select



Clark's select

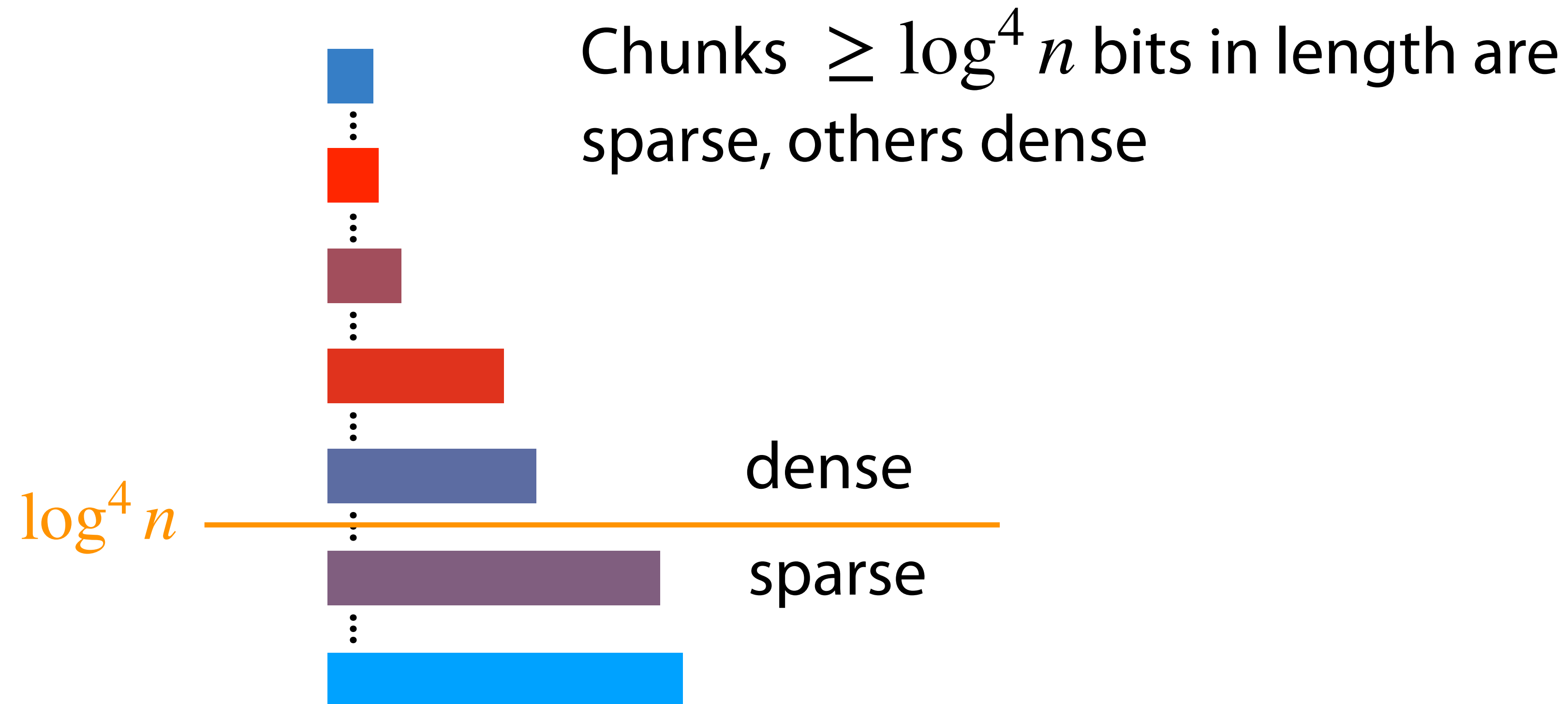


Clark's select



$\log^4 n$ is square of the # of set bits per chunk, $\log^2 n$

Clark's select



$\log^4 n$ is square of the # of set bits per chunk, $\log^2 n$

("sparse" roughly means "less than $\sqrt{\# \text{ bits are set}}$ ")

Clark's select: sparse case

Pre-calculate $B \cdot \text{select}_1$ for 1-bits in sparse chunks

Clark's select: sparse case

Pre-calculate B . select_1 for 1-bits in sparse chunks

$$O\left(\frac{n}{\log^4 n} \cdot \log n \cdot \log^2 n\right)$$

Clark's select: sparse case

Pre-calculate B . select₁ for 1-bits in sparse chunks

$$O \left(\underbrace{\frac{n}{\log^4 n}}_{\text{Max \# sparse chunks}} \cdot \log n \cdot \log^2 n \right)$$

Max # sparse chunks

Clark's select: sparse case

Pre-calculate B . select₁ for 1-bits in sparse chunks

$$O \left(\underbrace{\frac{n}{\log^4 n}}_{\text{Max \# sparse chunks}} \cdot \underbrace{\log n}_{\text{\# bits to store 1 answer}} \cdot \log^2 n \right)$$

Max # sparse chunks

bits to
store 1
answer

Clark's select: sparse case

Pre-calculate B . select₁ for 1-bits in sparse chunks

$$O \left(\underbrace{\frac{n}{\log^4 n}}_{\text{Max \# sparse chunks}} \cdot \underbrace{\log n}_{\substack{\text{\# bits to} \\ \text{store 1} \\ \text{answer}}} \cdot \underbrace{\log^2 n}_{\substack{\text{\# answers} \\ \text{per chunk}}} \right)$$

Clark's select: sparse case

Pre-calculate B . select₁ for 1-bits in sparse chunks

$$O \left(\underbrace{\frac{n}{\log^4 n}}_{\text{Max \# sparse chunks}} \cdot \underbrace{\log n}_{\substack{\text{\# bits to} \\ \text{store 1} \\ \text{answer}}} \cdot \underbrace{\log^2 n}_{\substack{\text{\# answers} \\ \text{per chunk}}} \right)$$

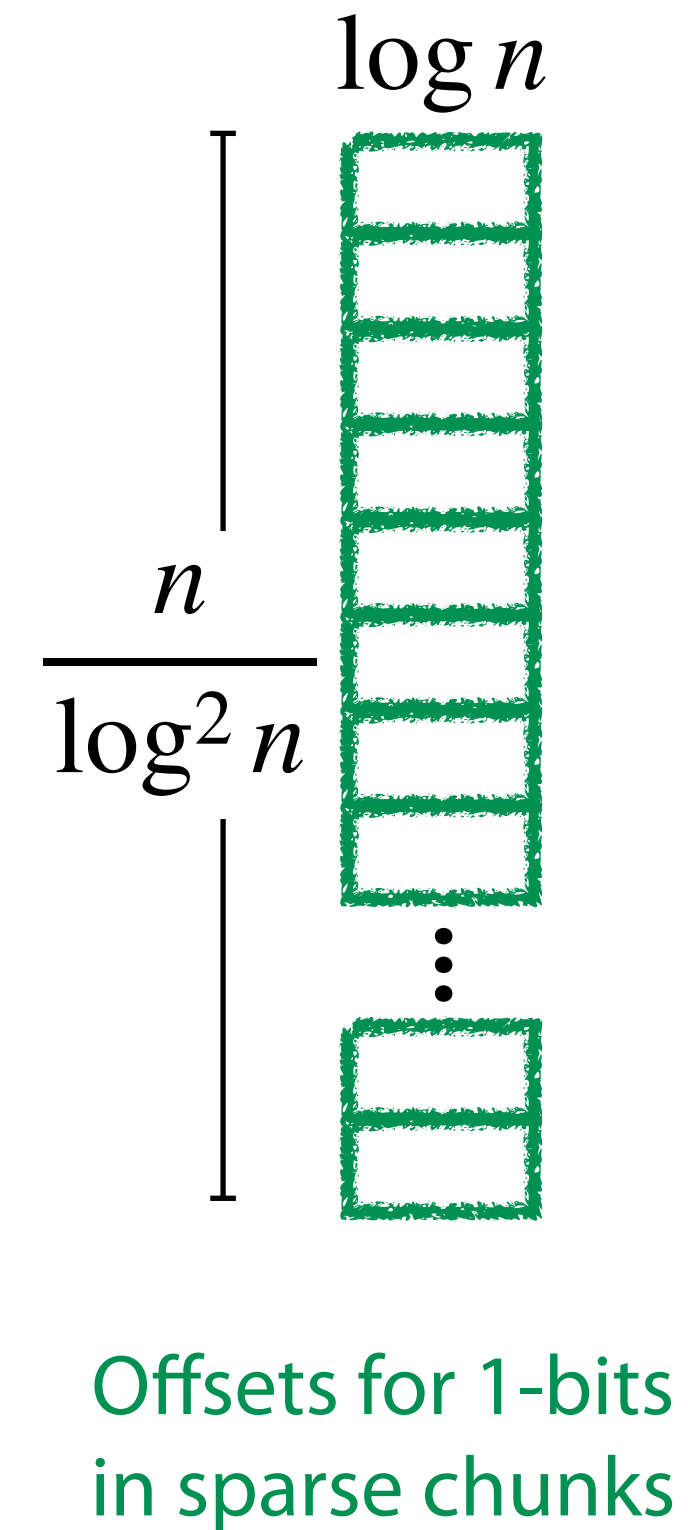
$$= O \left(\frac{n}{\log n} \right) = \check{O}(n)$$

Clark's select: sparse case

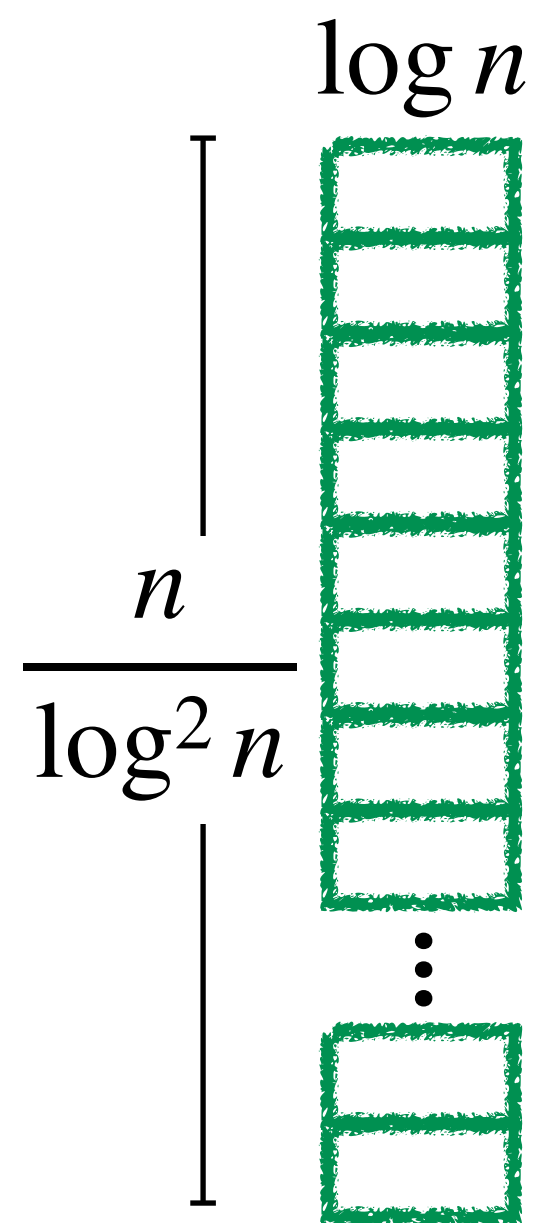
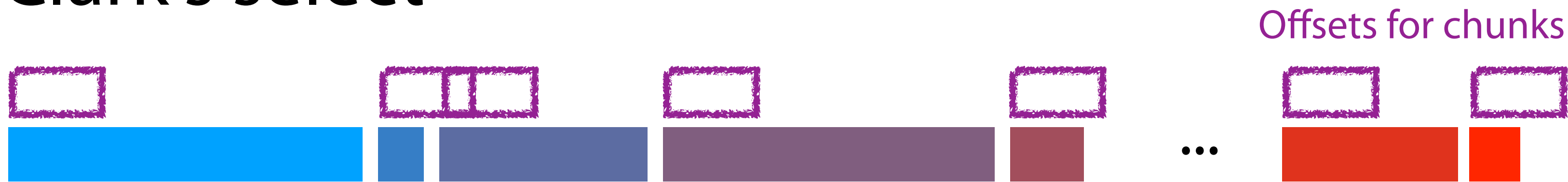
Pre-calculate B . select₁ for 1-bits in sparse chunks

$$O\left(\underbrace{\frac{n}{\log^4 n}}_{\text{Max \# sparse chunks}} \cdot \underbrace{\log n}_{\text{\# bits to store 1 answer}} \cdot \underbrace{\log^2 n}_{\text{\# answers per chunk}}\right)$$

$$= O\left(\frac{n}{\log n}\right) = \tilde{O}(n)$$



Clark's select

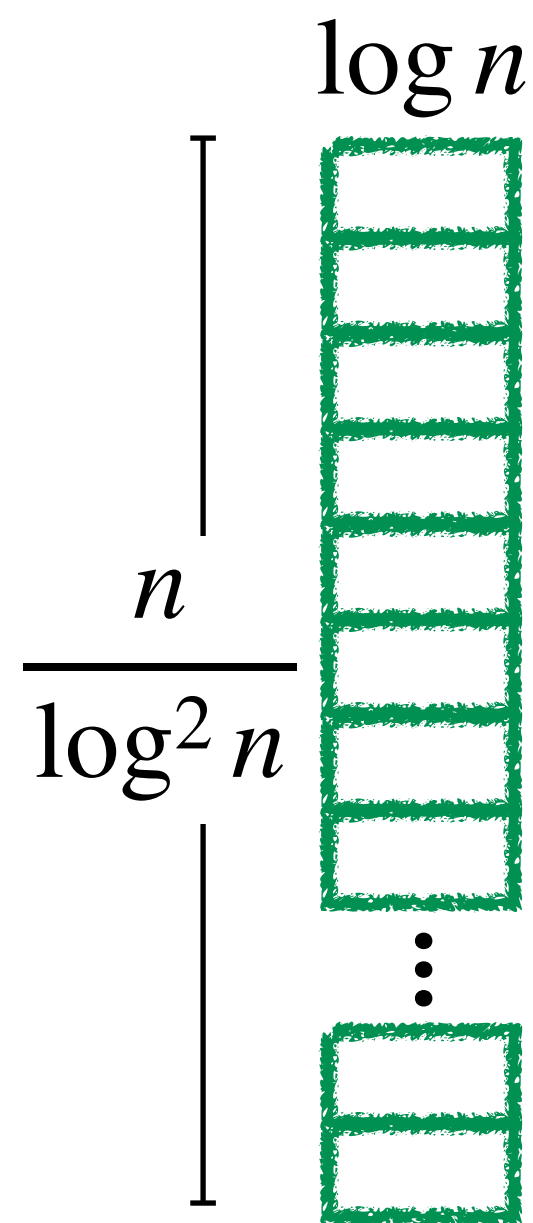


Offsets for 1-bits
in sparse chunks

Clark's select

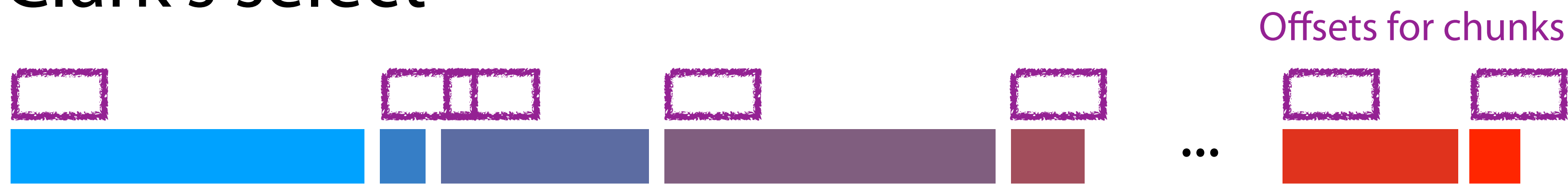


So far, strategy for select is:



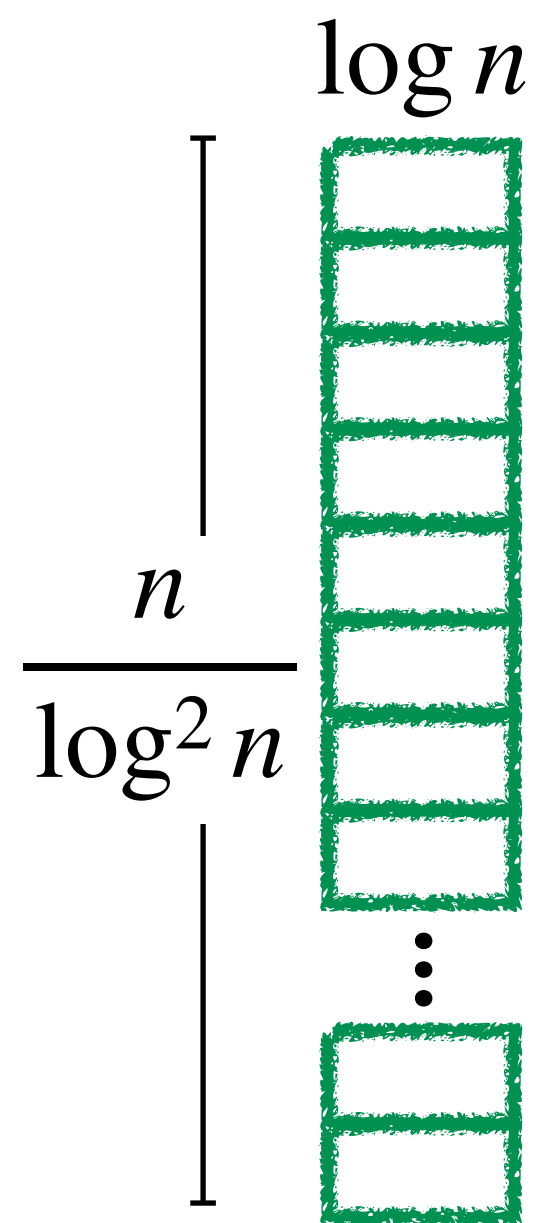
Offsets for 1-bits
in sparse chunks

Clark's select



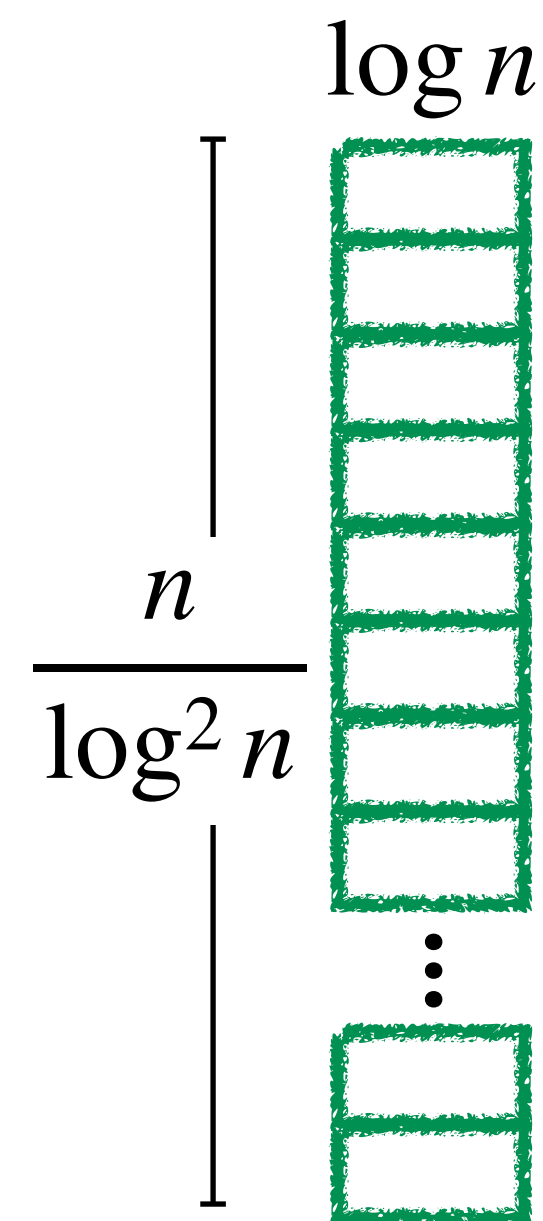
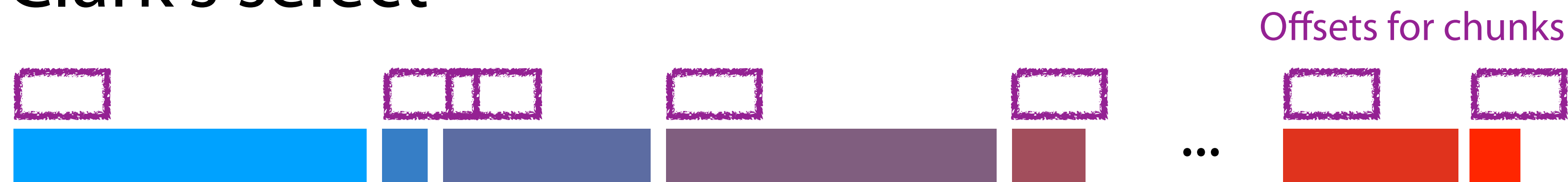
So far, strategy for select is:

(a) find what chunk it's in (division)



Offsets for 1-bits
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Clark's select



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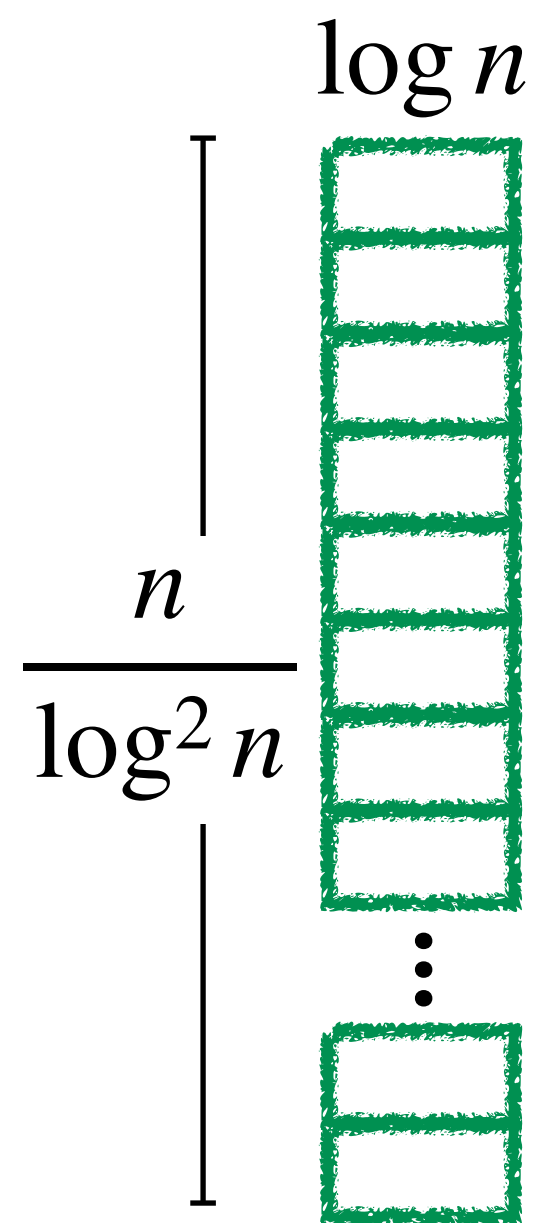
- (a) find what chunk it's in (division)
- (b) if chunk is sparse ($\geq \log^4 n$ bits)

Clark's select



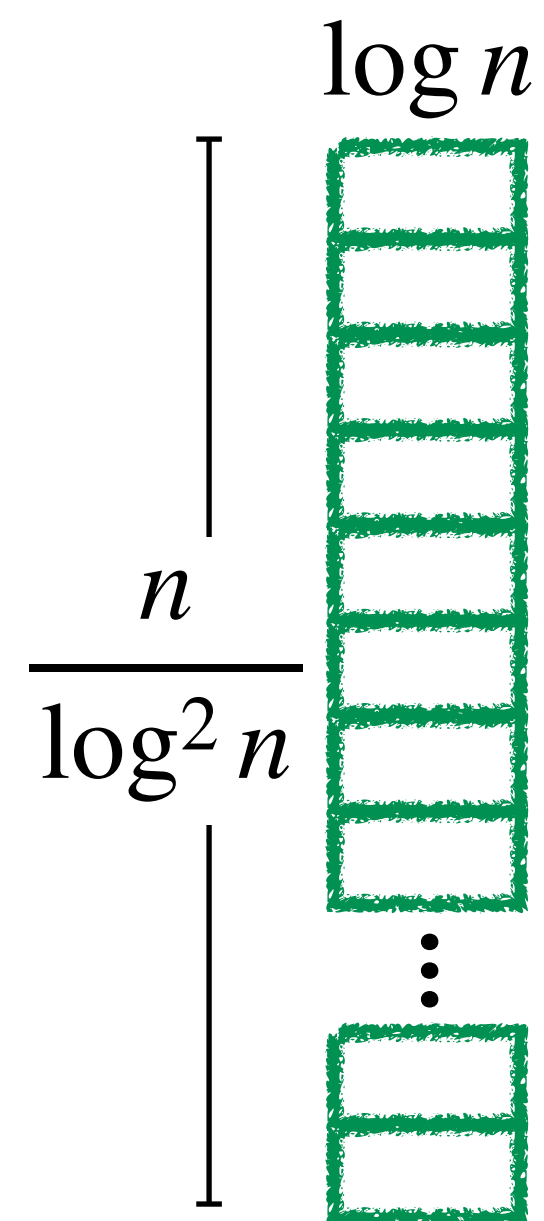
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- (a) find what chunk it's in (division)
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 - (b.i) look up in **sparse offset table**



Offsets for 1-bits
in sparse chunks

Clark's select

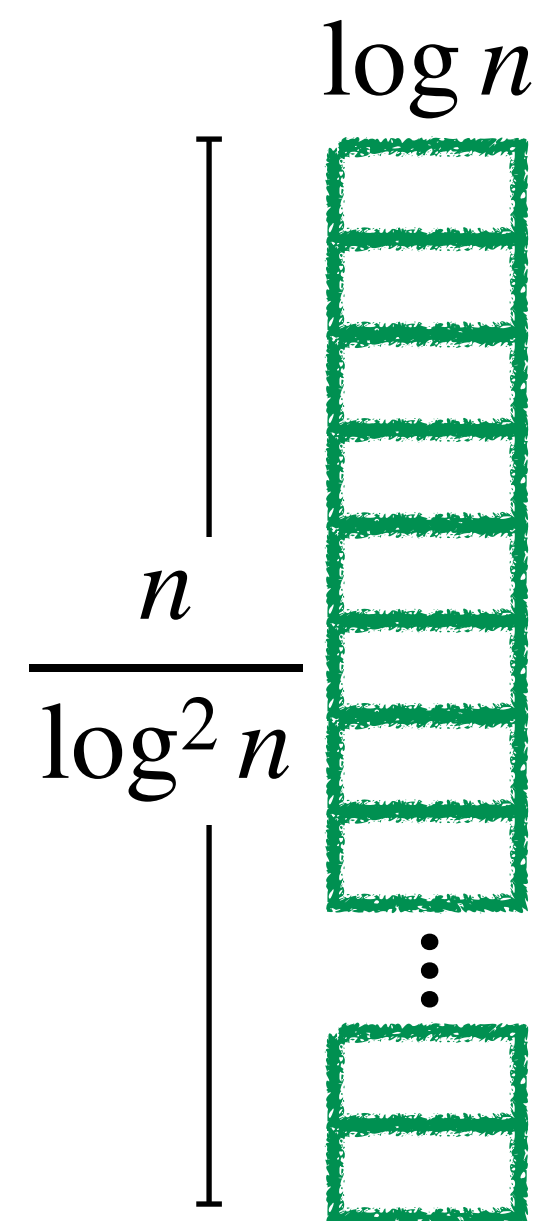


Offsets for 1-bits
in sparse chunks

So far, strategy for select is:

- (a) find what chunk it's in (division)
- (b) if chunk is sparse ($\geq \log^4 n$ bits)
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- (c) if chunk is dense ($< \log^4 n$ bits)

Clark's select



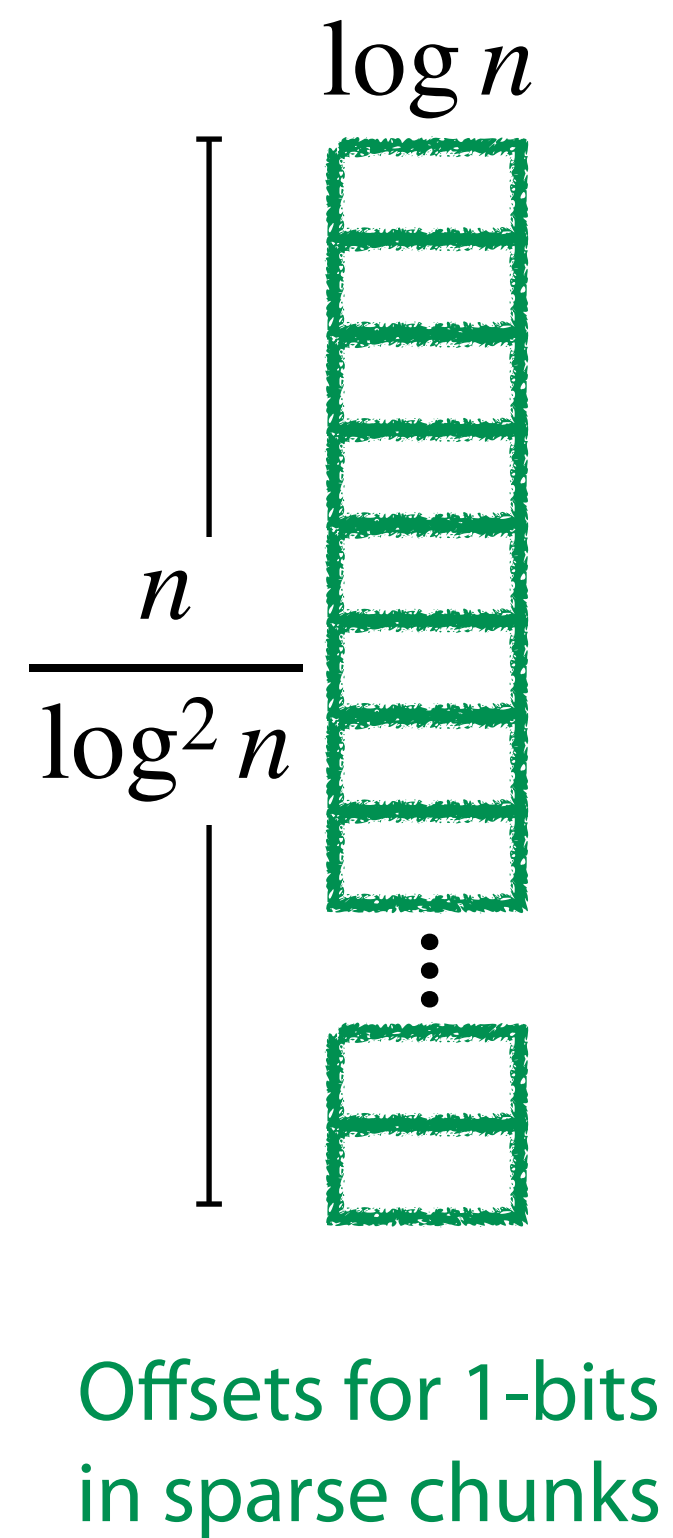
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TODO

Clark's select



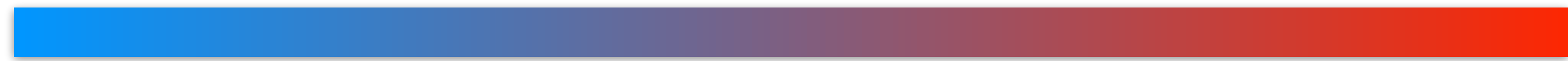
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TODO

So far, space is $\tilde{O}(n)$

Clark's select



Split into $\log^2 n$ -weight chunks



Sparse ($\geq \log^4 n$ -length)

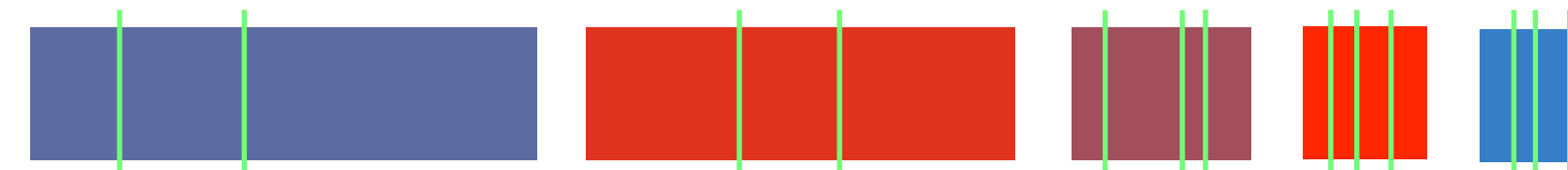


Lookup table for
each 1-bit ✓

Dense ($< \log^4 n$ -length)



Split into $\sqrt{\log n}$ -weight sub-chunks



Sparse ($\geq 1/2 \log n$ -length)



Lookup table for
each 1-bit ✓

Dense ($< 1/2 \log n$ -length)



Lookup table for all
possible sub-chunks ✓

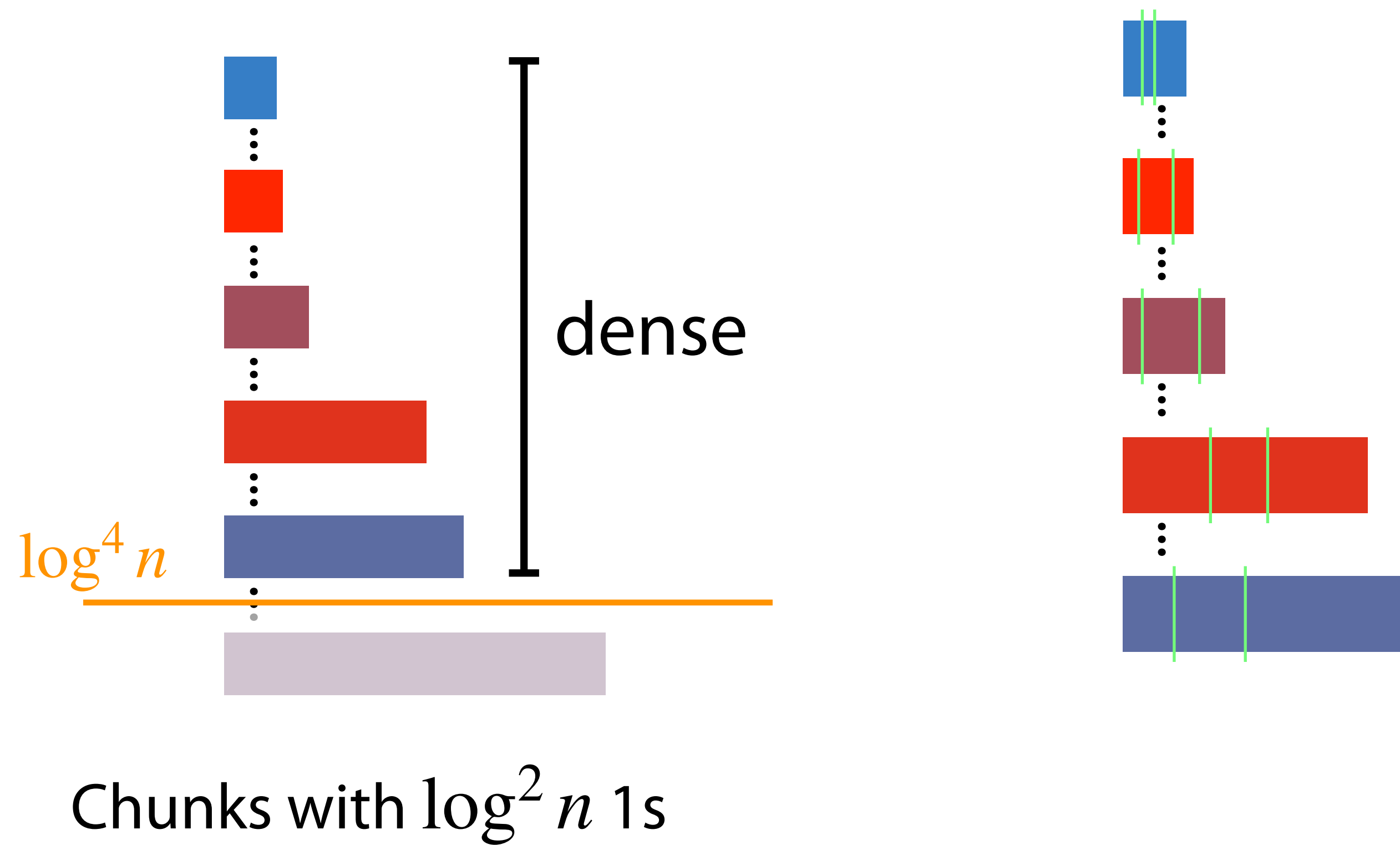
Clark's select: dense case

Dense chunks are shorter than $\log^4 n$ bits; further
subdivide these to sub-chunks of $\sqrt{\log n}$ 1-bits each



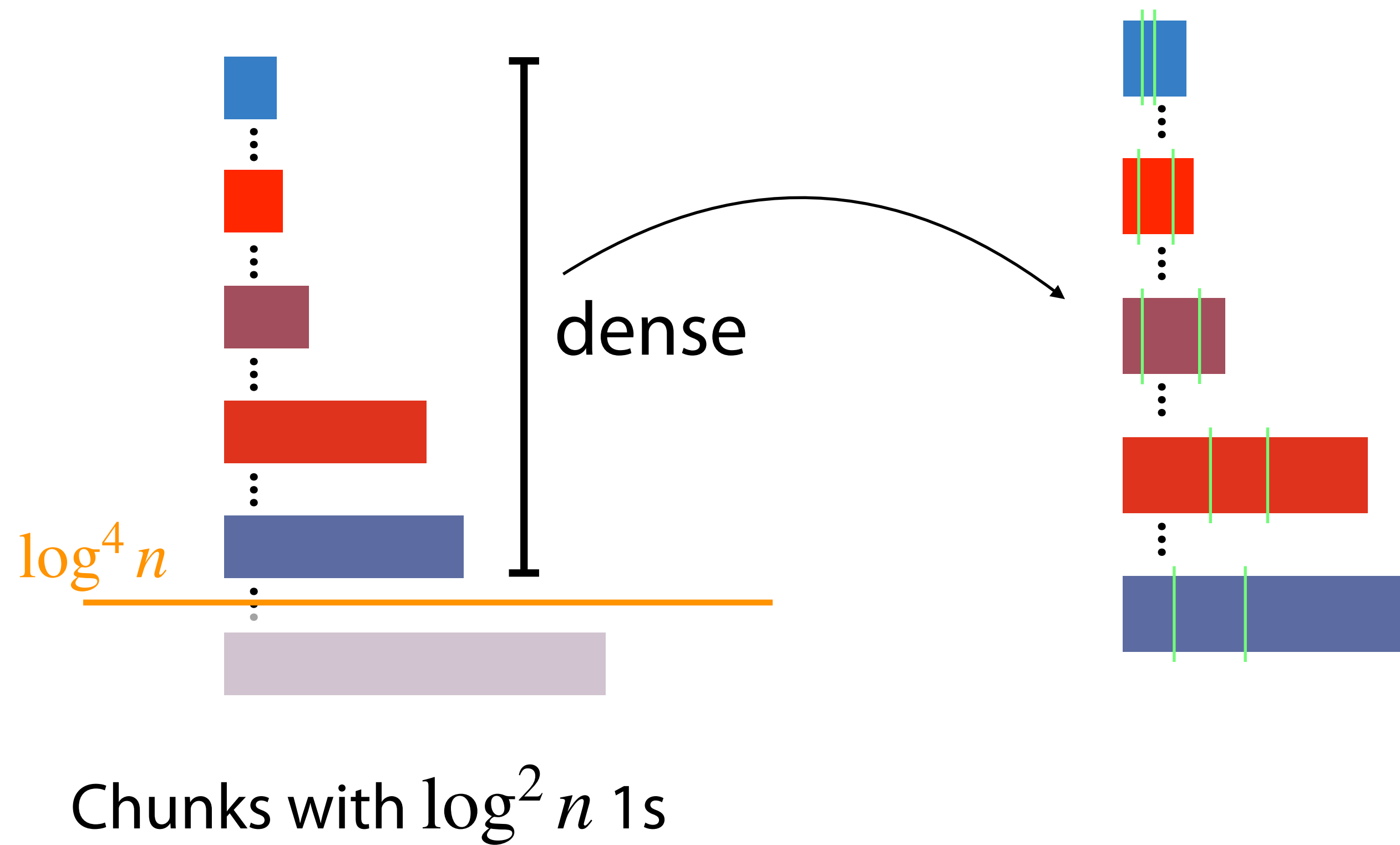
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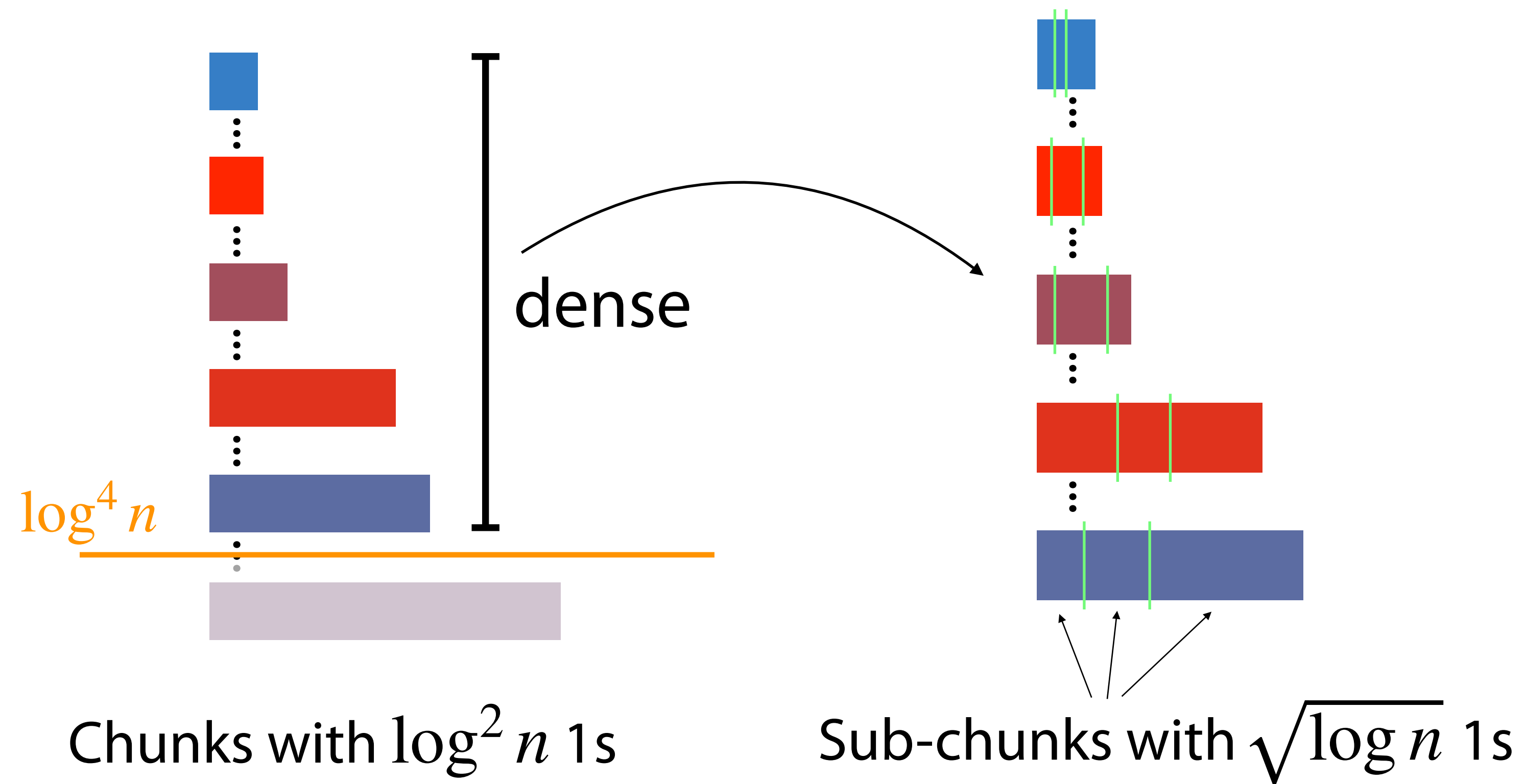
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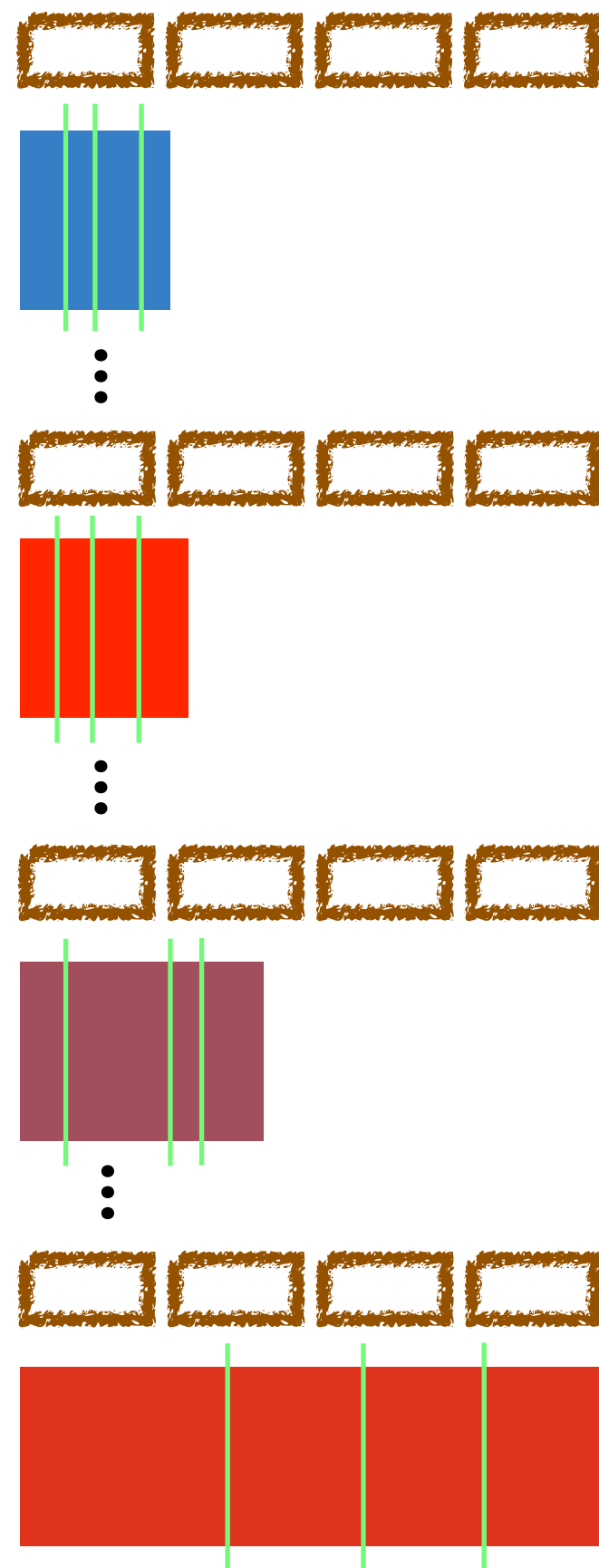


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Clark's select: dense case

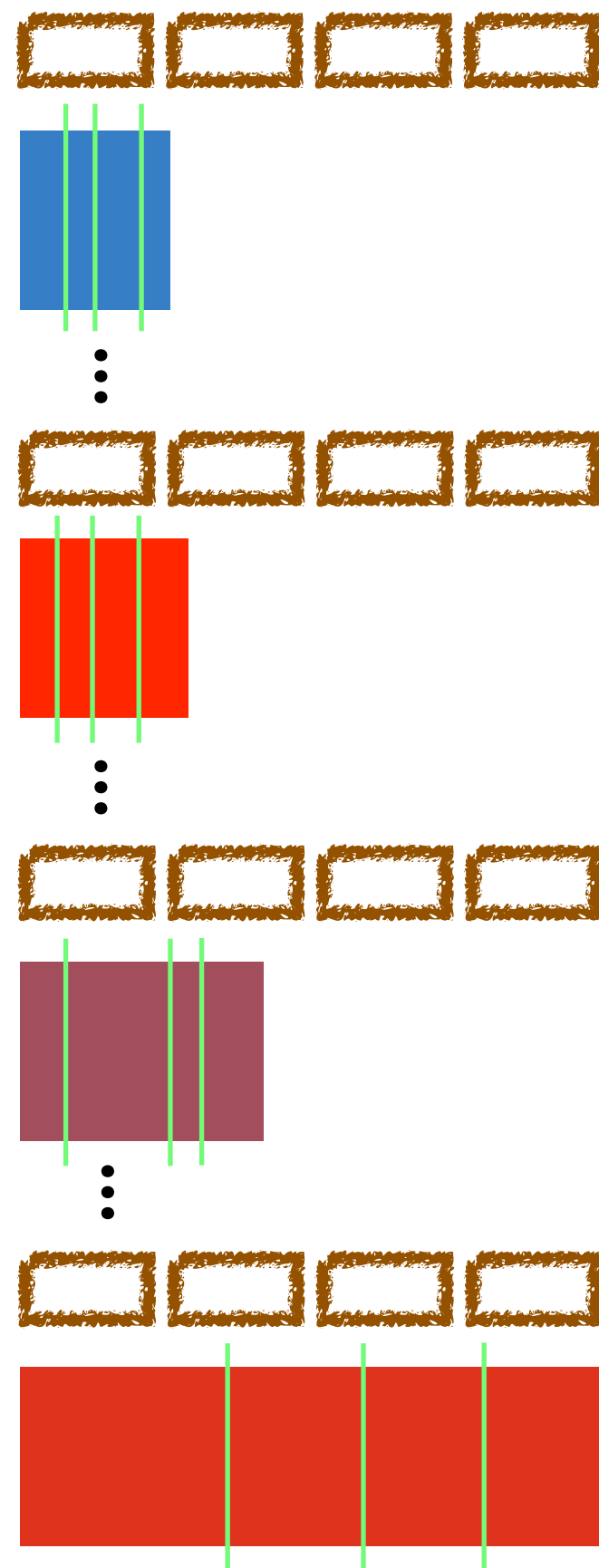


Store **relative offset** per sub-chunk

Containing chunk has $< \log^4 n$ bits,
so **relative offset** fits in

$$O(\log \log^4 n) = O(\log \log n) \text{ bits}$$

Clark's select: dense case



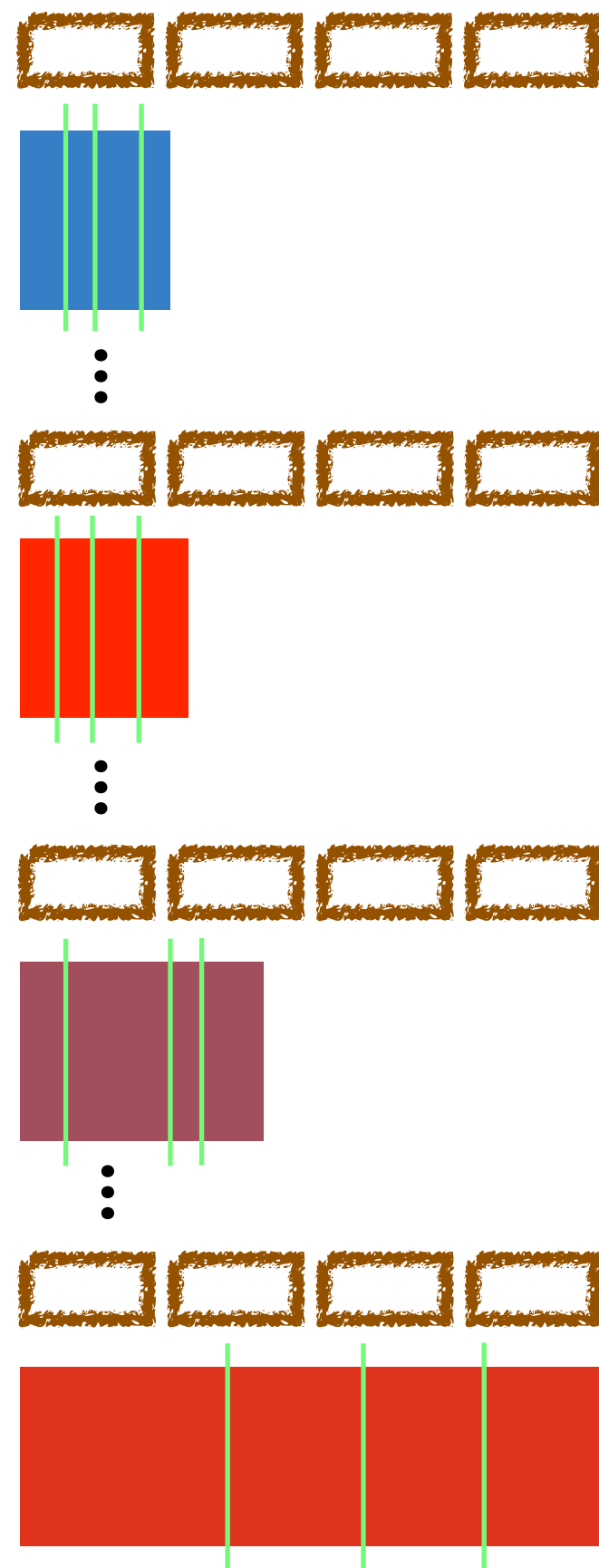
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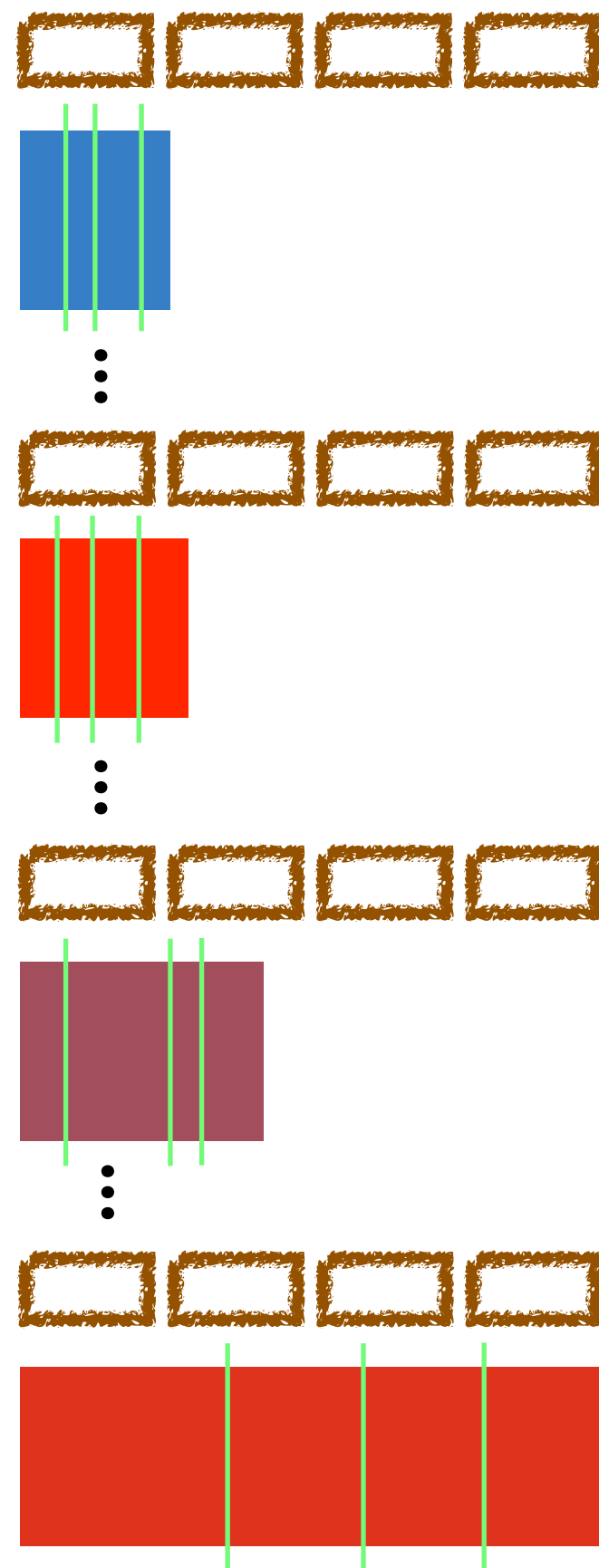
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Overall:

Clark's select: dense case



Store **relative offset** per sub-chunk

There are $\leq n/\sqrt{\log n}$ sub-chunks

Containing chunk has $< \log^4 n$ bits,
so **relative offset** fits in

$O(\log \log^4 n) = O(\log \log n)$ bits

Overall: $O\left(\frac{n \log \log n}{\sqrt{\log n}}\right) = \tilde{o}(n)$

Clark's select

So far, strategy for select is:

Clark's select

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- (a) find what chunk it's in (division)
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 - (c.ii) find what sub-chunk it's in (division by $\sqrt{\log n}$)

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 - (c.iii) look up sub-chunk's **relative offset**

TODO: need to look within sub-chunks

Clark's select: dense/sparse case

Sub-chunks with $\geq 1/2 \log n$ bits are sparse; we simply store relative offsets for every 1-bit

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Overall: $O\left(\frac{n}{1/2 \log n} \log \log n \sqrt{\log n}\right)$

Clark's select: dense/sparse case

Sub-chunks with $\geq 1/2 \log n$ bits are sparse; we simply store relative offsets for every 1-bit

Overall: $O \left(\underbrace{\frac{n}{1/2 \log n}}_{\text{Max \# sparse sub-chunks}} \log \log n \sqrt{\log n} \right)$

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Overall: $O \left(\underbrace{\frac{n}{1/2 \log n}}_{\text{Max \# sparse sub-chunks}} \underbrace{\log \log n}_{\substack{\text{\# bits to store} \\ \text{1 answer (rel.} \\ \text{to chunk)}}} \sqrt{\log n} \right)$

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Sub-chunks with $\geq 1/2 \log n$ bits are sparse; we simply store relative offsets for every 1-bit

Overall: $O \left(\underbrace{\frac{n}{1/2 \log n}}_{\text{Max \# sparse sub-chunks}} \underbrace{\log \log n}_{\text{\# bits to store 1 answer (rel. to chunk)}} \underbrace{\sqrt{\log n}}_{\text{\# 1-bits per chunk}} \right)$

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Sub-chunks with $\geq 1/2 \log n$ bits are sparse; we simply store relative offsets for every 1-bit

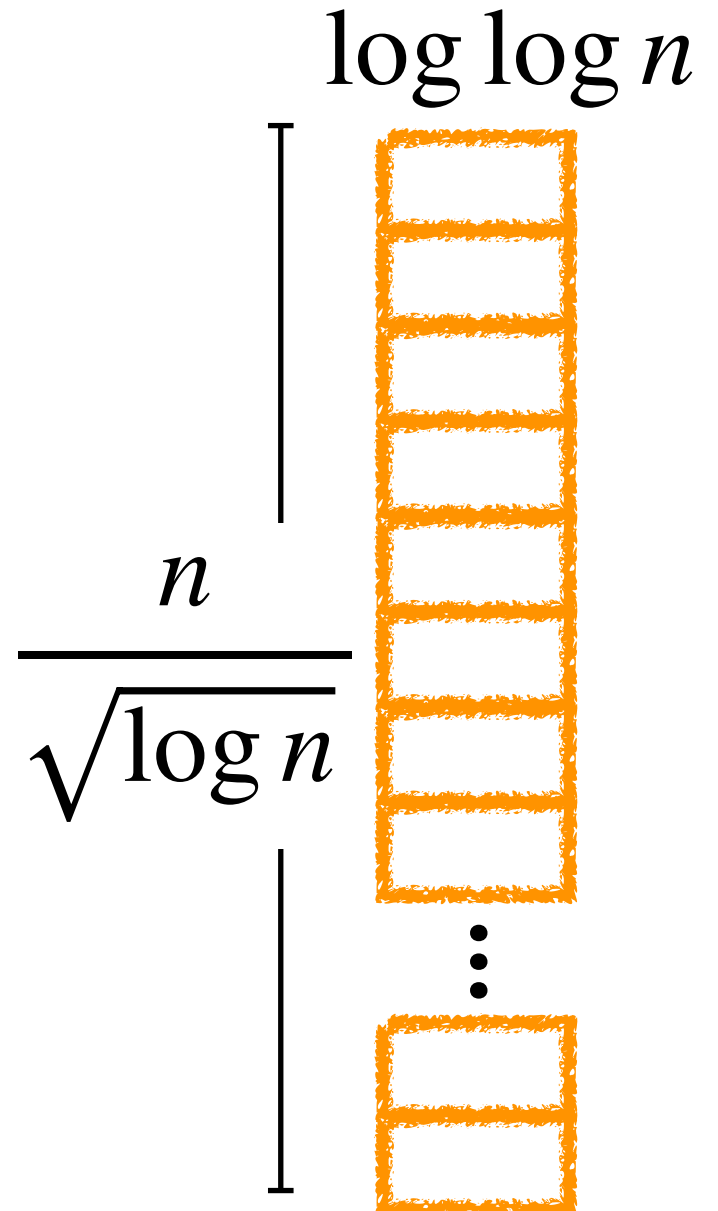
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$$= O \left(\frac{n \sqrt{\log n} \log \log n}{\log n} \right) = O \left(\frac{n \log \log n}{\sqrt{\log n}} \right) = \tilde{o}(n)$$

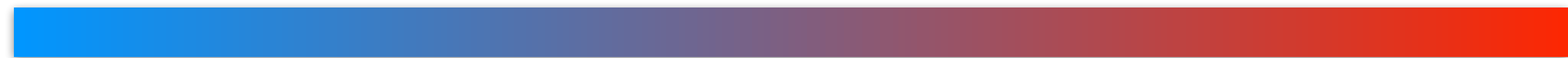
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Clark's select



Split into $\log^2 n$ -weight chunks



Sparse ($\geq \log^4 n$ -length)

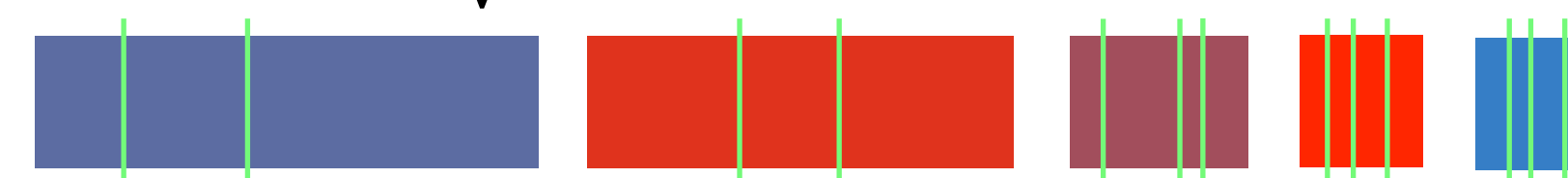


Lookup table for
each 1-bit ✓

Dense ($< \log^4 n$ -length)



Split into $\sqrt{\log n}$ -weight sub-chunks

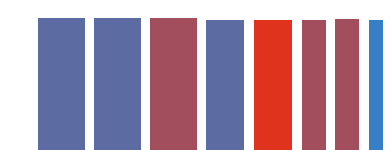


Sparse ($\geq 1/2 \log n$ -length)



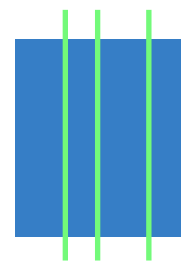
Lookup table for
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Dense ($< 1/2 \log n$ -length)

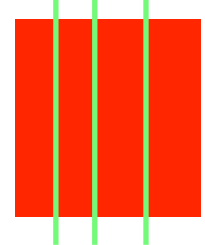


Lookup table for all
possible sub-chunks ✓

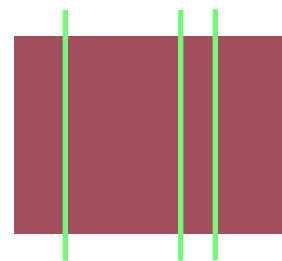
Clark's select: dense/dense case



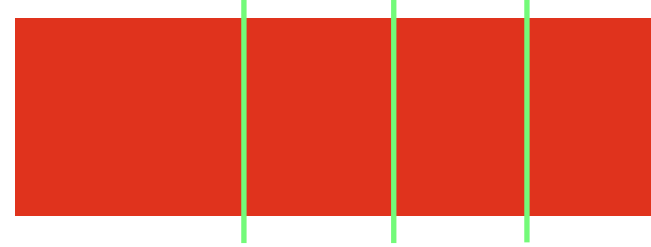
⋮



⋮



⋮



Sub-chunks $< 1/2 \log n$ bits are dense;
pre-calculate answers for all such chunks, like rank:

$$2^{1/2 \log n} \cdot \sqrt{\log n} \cdot \log \log n$$

possible
bitvectors

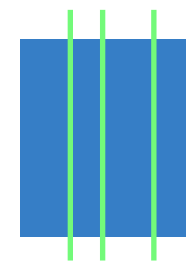
possible
1-bits

answer

$$= O\left(\sqrt{n \log n} \log \log n\right)$$

$$= \tilde{o}(n)$$

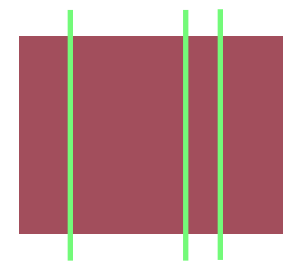
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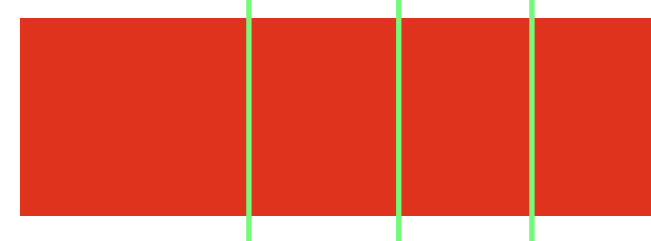
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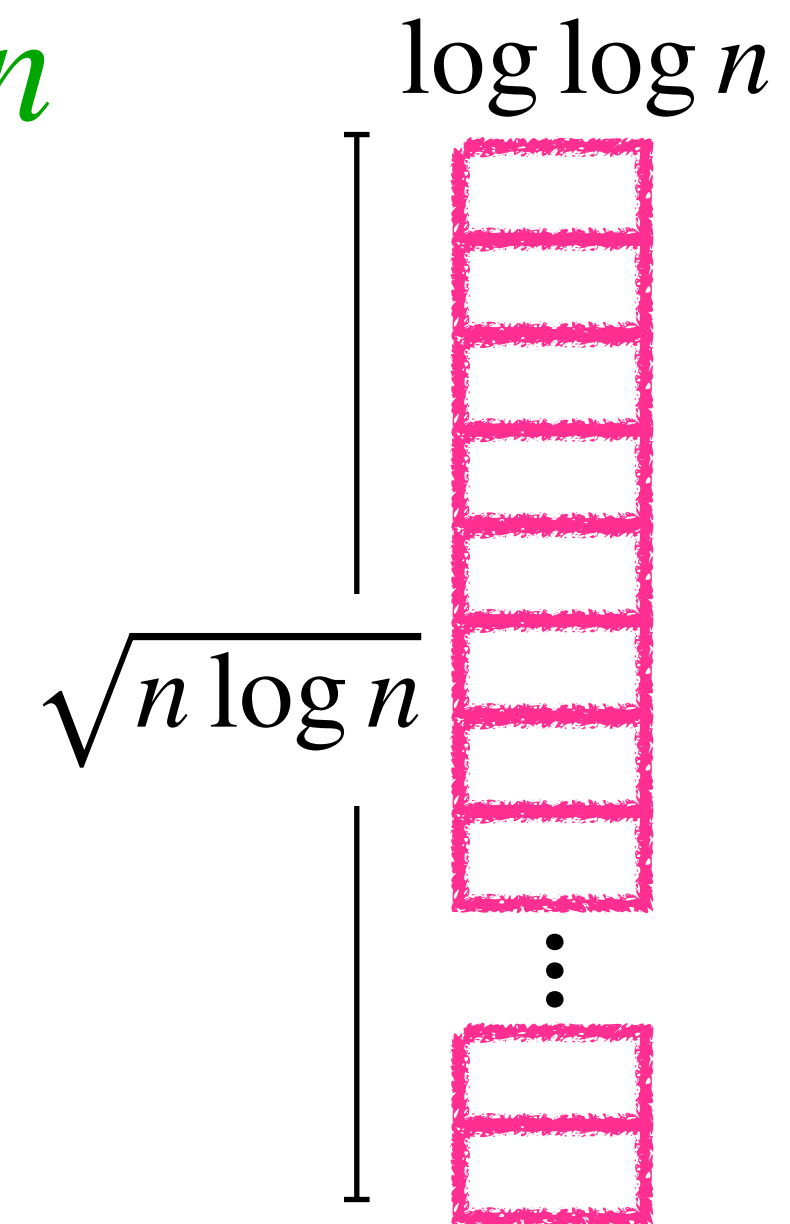
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$$= O\left(\sqrt{n \log n} \log \log n\right)$$

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Clark's select

- (a) find what chunk it's in (division by $\log^2 n$)
- (b) if chunk is sparse ($\geq \log^4 n$ bits)
- (c) if chunk is dense ($< \log^4 n$ bits)

Clark's select

- (a) find what chunk it's in (division by $\log^2 n$)
- (b) if chunk is sparse ($\geq \log^4 n$ bits)
 - (b.i) look up answer in **sparse offset table**
- (c) if chunk is dense ($< \log^4 n$ bits)

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- (a) find what chunk it's in (division by $\log^2 n$)
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 - (c.i) look up **chunk's offset**
 - (c.ii) find what sub-chunk it's in (divide by $\sqrt{\log n}$)
 - (c.iii) look up sub-chunk's **relative offset**
 - (c.iv) if sub-chunk is sparse ($\geq 1/2 \log n$ bits)
 - (c.v) if sub-chunk is dense

Clark's select

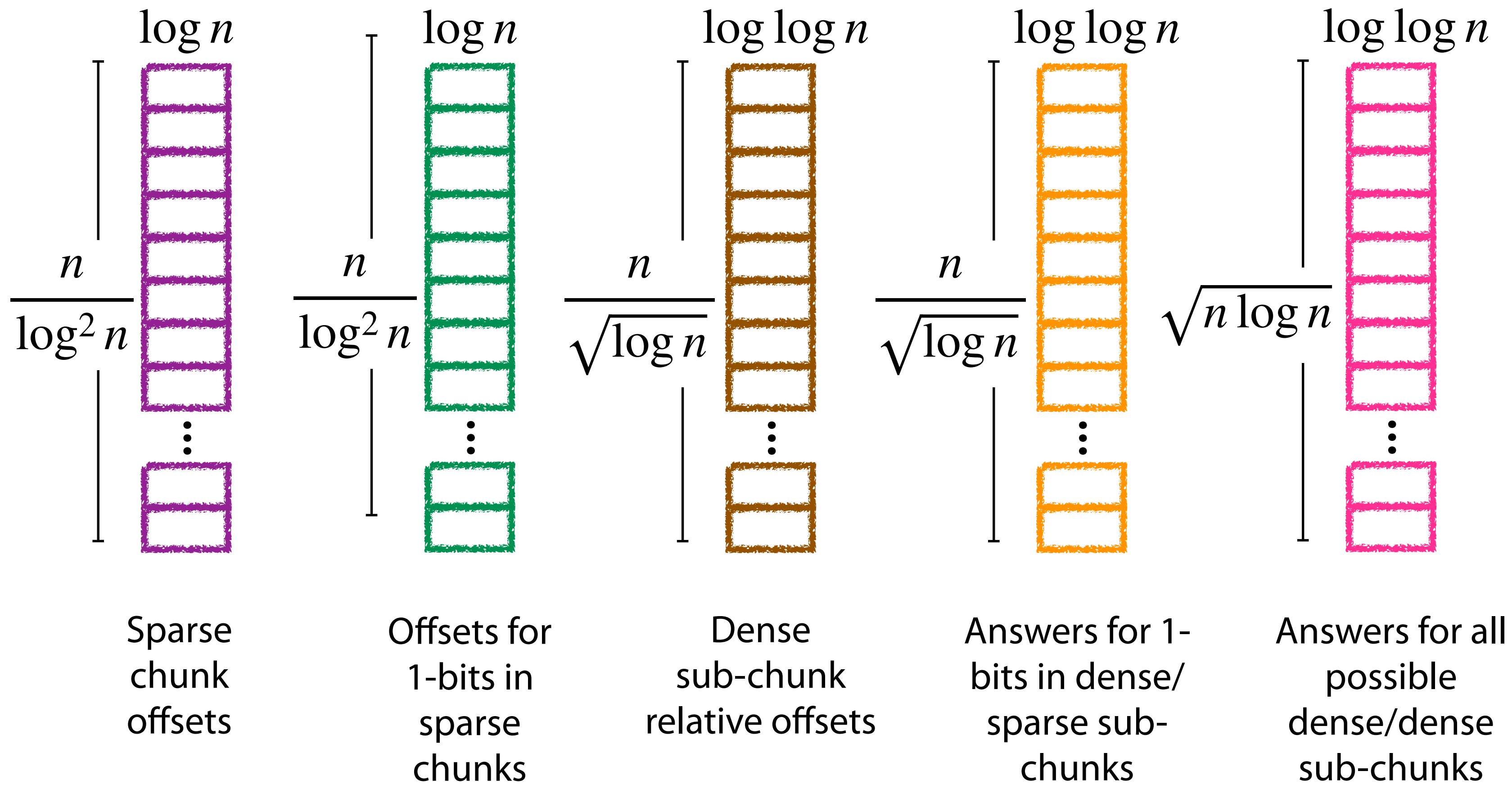
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 - (c.ii) find what sub-chunk it's in (divide by $\sqrt{\log n}$)
 - (c.iii) look up sub-chunk's **relative offset**
 - (c.iv) if sub-chunk is sparse ($\geq 1/2 \log n$ bits)
 - (c.iv.A) look up answer in **sparse 1-bit table**
 - (c.iv.B) return (c.i) + (c.iii) + (c.iv.A)
 - (c.v) if sub-chunk is dense

Clark's select

- (a) find what chunk it's in (division by $\log^2 n$)
- (b) if chunk is sparse ($\geq \log^4 n$ bits)
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 - (c.i) look up **chunk's offset**
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 - (c.iii) look up sub-chunk's **relative offset**
 - (c.iv) if sub-chunk is sparse ($\geq 1/2 \log n$ bits)
 - (c.iv.A) look up answer in **sparse 1-bit table**
 - (c.iv.B) return (c.i) + (c.iii) + (c.iv.A)
 - (c.v) if sub-chunk is dense
 - (c.v.A) look up answer in **all possible dense/dense table**
 - (c.v.B) return (c.i) + (c.iii) + (c.v.A)

Clark's select

Overall, space is $\tilde{O}(n)$



Bitvectors

	Time	Space (bits)	Note
$B . \text{access}$	$O(1)$	n	Lookup
$B . \text{select}_1$	$O(1)$	$\tilde{O}(n)$	 Clark
$B . \text{rank}_1$	$O(1)$	$\tilde{O}(n)$	 Jacobson